

Q1.

UNIT-3: GRAPH THEORYDefinition:- GRAPH:

An undirected graph G is an ordered triple $(V(G), E(G), \phi)$, consists of a non-empty set $V(G)$, called the set of vertices, (nodes, points) of the Graph. $E(G)$ said to be set of edges of the graph and ϕ is a mapping from the set of edges E to set of order or unordered pairs of elements of $V(G)$.

EX:-

$$G = (V(G), E(G), \phi)$$

$$\text{where } V(G) = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

$$E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

Q2.

$$\text{and } \phi(e_1) = (v_1, v_2)$$

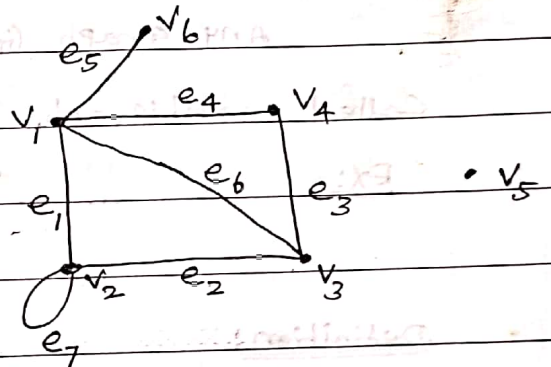
$$\phi(e_2) = (v_2, v_3)$$

$$\phi(e_3) = (v_3, v_4)$$

$$\phi(e_4) = (v_4, v_1)$$

$$\phi(e_5) = (v_1, v_6)$$

$$\phi(e_6) = (v_1, v_3) \quad \phi(e_7) = (v_2, v_3) \quad (\text{or}) \quad \phi(e_7) = (v_3, v_2)$$

Definition: Adjacent vertices:-

Any pair of vertices which are connected by an edge in a graph is called adjacent vertices. EX: $v_1 \xrightarrow{e_1} v_2$

v_1 & v_2 are adjacent vertices.

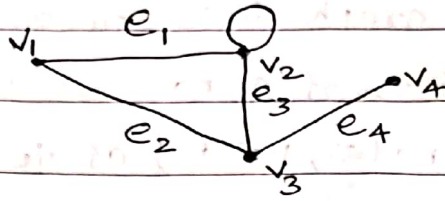
Definition: loop:-

An edge of a graph which joins a vertex to itself is called a loop. EX:- $v_1 \xrightarrow{e_2} v_1$ Hence the edge e_2 is a loop.

Definition:

Q3.

The degree of the vertex V in an undirected graph is the number of edges incident with V . It is denoted that $d(V)$.

Ex:-

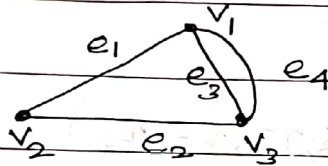
$$d(v_1) = 2 : d(v_3) = 3$$

$$d(v_2) = 4 ; d(v_4) = 1.$$

$$1+1+2$$

Definition:-

In a graph G , if same pairs of vertices are joined by more than one edge. Such edges are called Parallel edges.

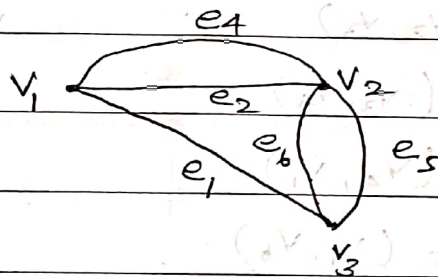
Ex:-

In this graph e_3 and e_4 are parallel edges.

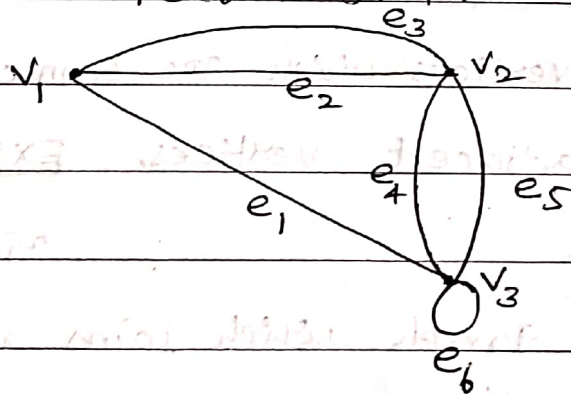
Definition:

Q4.

Any graph G which contain some parallel edges is called multigraph.

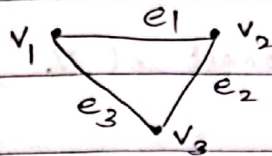
Ex:-Definition:

Any graph G which has loops and multiple or Parallel edges is called pseudo graph.



Definition:

Q5. Any graph G_1 , has no parallel edges and loops is called a simple graph.

Definition:-

A graph unit with one vertex is said to be a trivial graph.

Definition:-

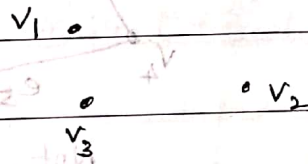
In a graph, a vertex is not adjacent to any other vertex is called an isolated vertex. That is $d(v) = 0$.

Ex:-

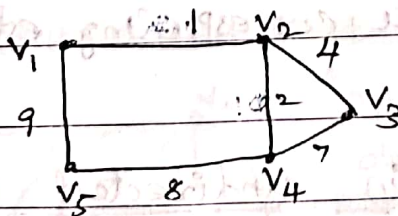
Hence v_4 is the isolated vertex.

Definition:

Q6. A graph with only isolated vertices is called a null graph that is $E(G)$ is empty and $V(G)$ is non-empty.

Ex:-Definition:

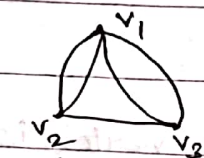
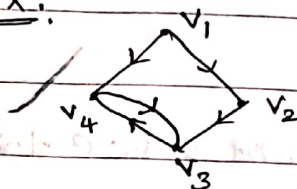
Any graph G in which weights are assigned to every edge is called a weighted graph.



Hence 1, 2, 4, 7, 8, 9 are weights assigned to each edges.

Definition: Digraph:

Q7. A digraph (or) a directed graph $G = (V, E)$ consists of a non-empty set of vertices V and directed edges E , which are ordered pairs of element of V .

EX:

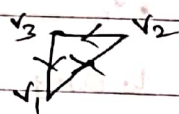
undirected multigraph



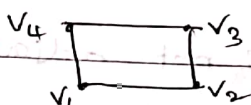
Directed multigraph



Directed graph with distinct edges.



Directed graph

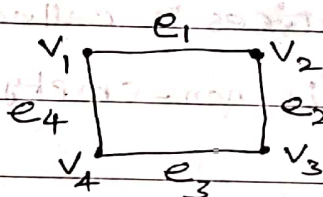


Undirected graph

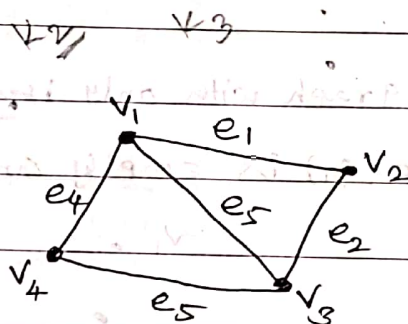
Definition:

A graph as which each vertex has K -degree is called K -regular graph.

Q8.

EX:

Regular graph



Not a regular graph

Definition: Pendent Vertex:

Any vertex having degree one is called a pendent vertex, and the corresponding edge is called pendent edge.

Degree of a vertex in undirected graph:-

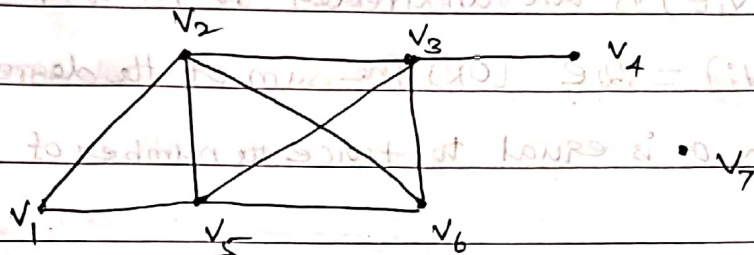
The degree of a vertex in an undirected graph is

the number of edges incident with it. It is denoted by $\deg(V)$.

Note: (1) A loop at a vertex contributes twice to the degree of that vertex.

(2) The degree of isolated vertex is zero.

Ex:-



$\deg(V_1) = 3$; $\deg(V_2) = 4$; $\deg(V_3) = 4$; $\deg(V_4) = 1 \rightarrow$ Pendent vertex.

$\deg(V_5) = 4$; $\deg(V_6) = 3$; $\deg(V_7) = 0 \rightarrow$ Isolated vertex.

Degree of a vertex in directed Graph:

Q 10.

Indegree:

In the digraph, the indegree of a vertex V is the number of edges with V as their terminal vertex. It is denoted by $\deg^-(V)$.

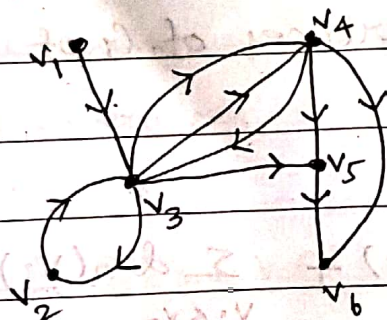
Out degree:

In the digraph, the outdegree of a vertex V is the number of edges with V as their initial vertex. It is denoted by $\deg^+(V)$.

Note:-

Loop will contribute to both indegree and outdegree.

Ex:-



Indegree

$$\deg^-(V_1) = 0$$

$$\deg^-(V_2) = 1$$

$$\deg^-(V_3) = 3$$

$$\deg^-(V_4) = 2$$

$$\deg^-(V_5) = 2$$

$$\deg^-(V_6) = 2$$

$$\sum \deg^-(V) : \text{Total} : 10$$

Outdegree

$$\deg^+(V_1) = 1$$

$$\deg^+(V_2) = 1$$

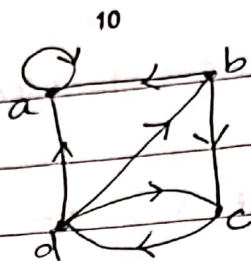
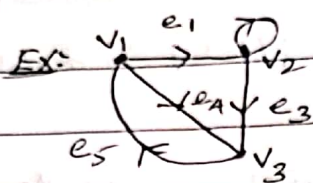
$$\deg^+(V_3) = 4$$

$$\deg^+(V_4) = 3$$

$$\deg^+(V_5) = 1$$

$$\deg^+(V_6) = 0$$

$$\sum \deg^+(V) : 10$$



Since

Theorem:

[The Handshaking Theorem]

If $G=(V,E)$ is an undirected graph with e edges, then $\sum \deg(v_i) = 2e$ [OR] The sum of the degree of all vertices in a graph is equal to twice the number of edges.

Proof:-

Since each edge has two end points. So each edge contribute two to the degrees.

Hence the sum of the degrees of all vertices in G is equal to twice the number of edges of G .

$$\text{i.e., } \sum_{i=1} \deg(v_i) = 2e = 2|E|.$$

Hence the proof.

Theorem:

The number of vertices of odd degree in an undirected graph is even. [OR] If $G=(V,E)$ is a directed graph with e edges then

Proof:-

Let $G=(V,E)$ be an undirected graph.

Let V_1 & V_2 be the sets of vertices of G of even and odd degrees respectively.

W.K.T By the theorem

$$2e = \sum_{v_i \in V_1} \deg(v_i) + \sum_{v_j \in V_2} \deg(v_j)$$

Since each $\deg(v_i)$ is even

$\Rightarrow \sum \deg(v_i)$ is even. EX:

$$2e = \sum_{v_i \in V_1} \deg(v_i) + \sum_{v_j \in V_2} \deg(v_j) \quad 1=N \quad \square$$

(even degree) = even + even.

The number of vertices of odd degree is even.

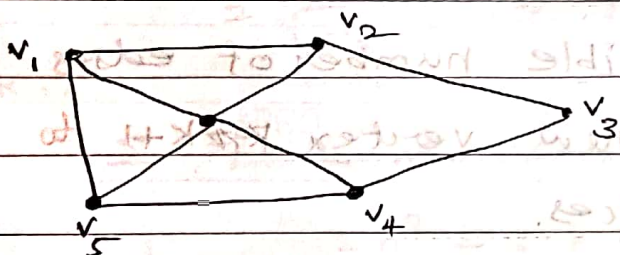
Hence the Proof.

Theorem:-

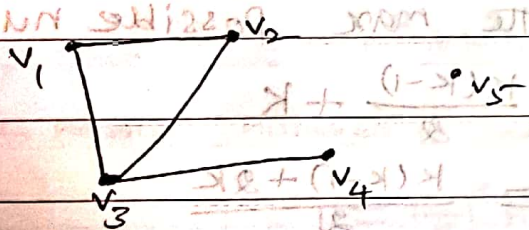
Definition: Degree sequence of a graph:

The degrees of the vertices in a graph G as a non-decreasing sequence then it's called degree sequence of a graph.

EX:-



The degree sequence of Graph is $0, 3, 3, 3, 3, 4$.



The degree sequence of Graph is $0, 1, 2, 2, 3$.

Theorem:-

The maximum number of edges in a simple graph with n -vertices is $\frac{n(n-1)}{2}$.

Proof:-

Let $G = (V, E)$ be any simple graph with n -vertices.

To prove:

The maximum number of edges in G is $\frac{n(n-1)}{2}$

We prove this by using induction on n .

Let $n=1$

Then G must be a graph with one vertex. Hence G is a simple graph and it must be an isolated vertex. Hence the maximum number of edges is zero

$$\text{i.e.) } \frac{1(1-1)}{2} = 0$$

Hence the result is true for $n=1$.

Now assume that the result is true for $n=k$

Hence a simple graph with ' k ' vertices and has the maximum number of edges $\frac{k(k-1)}{2}$

Now we prove the result is true for $n=k+1$

Now construct a new simple graph of ' k ' vertices with maximum possible number of edges $\frac{k(k-1)}{2}$.

Now introduce a new vertex $k+1$ to the remaining k vertices.

For this graph the max possible number of edges is

$$= \frac{k(k-1)}{2} + k$$

$$= \frac{k(k-1) + 2k}{2}$$

$$= \frac{k^2 - k + 2k}{2}$$

$$= \frac{k^2 + k}{2}$$

$$= \frac{k(k+1)}{2}$$

$$= \frac{(k+1)((k+1)-1)}{2}$$

Assumption
 $\left[\frac{n(n-1)}{2} \right]$

Hence the result is true for $k+1$.

$\therefore$ The result is true for any n .

Hence the proof.

Theorem:

Show that a simple graph with n -vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges.

Sol:

Let k components hence n_j vertices where $j=1, 2, \dots, k$ and $n_1 + n_2 + \dots + n_k = n$. (or) $\sum_{i=1}^k n_i = n$.

$\therefore$ The maximum number of edges in all the components is $= \frac{1}{2} \sum_{i=1}^k n_i(n_i - 1)$ [By the previous theorem].

$$= \frac{1}{2} \sum_{i=1}^k (n_i^2 - n_i) = \frac{1}{2} \sum_{i=1}^k n_i^2 - \frac{1}{2} \sum_{i=1}^k n_i = \frac{1}{2} \sum_{i=1}^k n_i^2 - \frac{1}{2} n$$

W.K.T

$$\sum n_i^2 \leq n^2 - (k-1)(2n-k)$$

$$= n^2 - 2nk + k^2 + 2n - k$$

$$= n^2 + k^2 - 2nk + 2n - k$$

$$\text{The max number of edges} \leq \frac{1}{2} [(n^2 + k^2 - 2nk + 2n - k) - n]$$

$$= \frac{1}{2} [n^2 + k^2 - 2nk + n - k]$$

$$= \frac{1}{2} (n-k)(n-k+1)$$

$$\text{The max number of edges} \leq \frac{1}{2} (n-k)(n-k+1)$$

Hence the theorem.

Some SPECIAL SIMPLE GRAPHS.

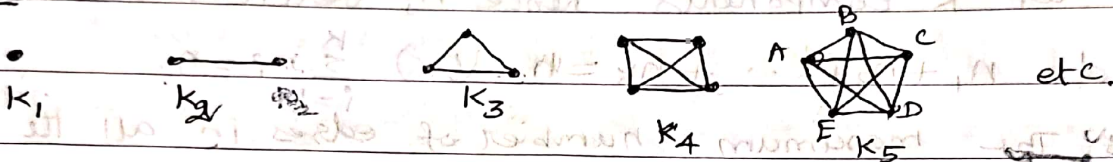
Complete graph:

A simple graph in which there is exactly one edge between each pair of distinct vertices is called a complete graph.

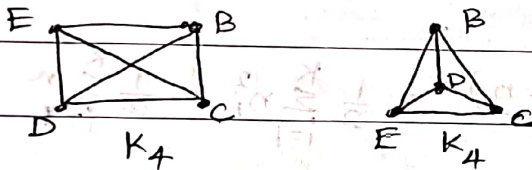
Note: The degree every vertex in K_n is $n-1$.

- ① Draw the complete graph K_5 with vertices A, B, C, D & E.
Draw all complete subgraph of K_5 with 4 vertices.

Sol:-



Complete subgraph with 4-vertices are



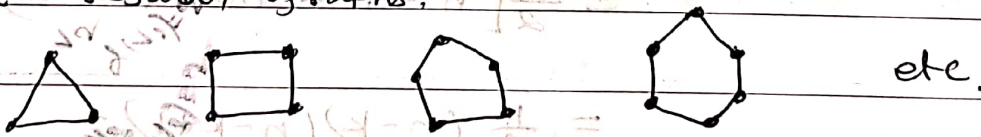
- ① Draw all the subgraphs of K_3 containing at least one vertex.

Regular Graph:

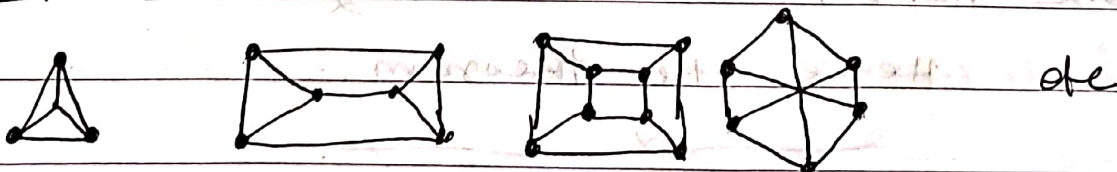
If every vertex of a simple graph has the same degree, then the graph is called a regular graph.

Note: The degree every vertex in K_n is $n-1$.

EX: 2-regular graphs:

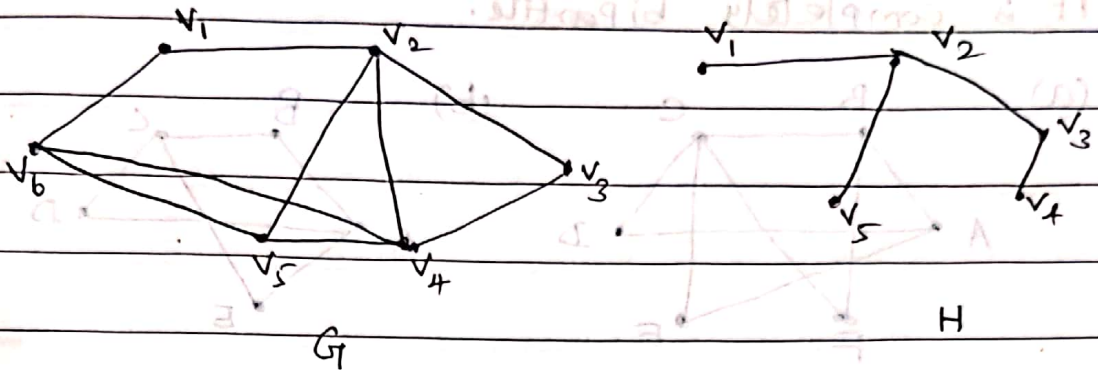


EX: 3-regular graphs:



Subgraphs:

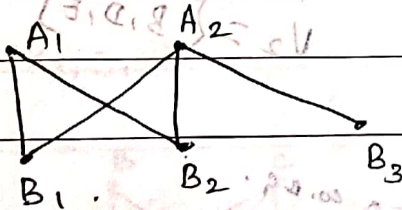
A graph $H = (V', E')$ is called a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

Ex:Bipartite graph:

If the vertex set V of a simple graph $G = (V, E)$ can be partitioned into two subsets V_1 and V_2 such that every edge of G connects a vertex in V_1 and a vertex in V_2 then G is called a Bipartite graph.

Note:

No edge in G connects either two vertices in V_1 or two vertices in V_2 .

Ex:

$$V_1 = \{A_1, A_2\}$$

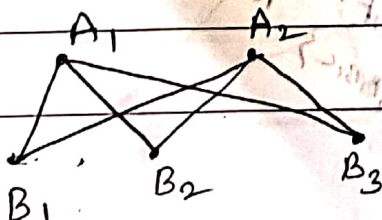
$$V_2 = \{B_1, B_2, B_3\}$$

Completely Bipartite graph:

If each vertex of V_1 is connected with every vertex of V_2 by an edge, then G is called a completely bipartite graph.

Ex:

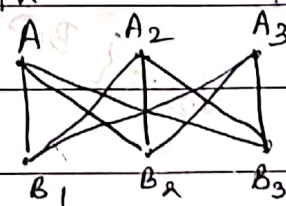
$K_{2,3}$
graph



$$V_1 = \{A_1, A_2\}$$

$$V_2 = \{B_1, B_2, B_3\}$$

$K_{3,3}$
graph



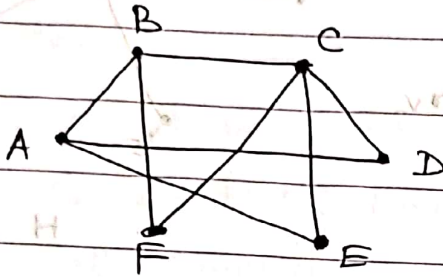
$$V_1 = \{A_1, A_2, A_3\}$$

$$V_2 = \{B_1, B_2, B_3\}$$

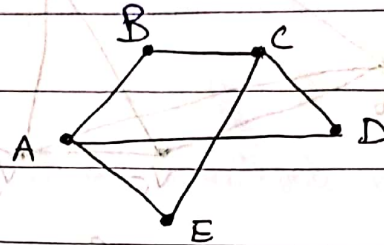
Ex:

Determine which of the following graphs are bipartite and which are not. If a graph is bipartite, state if it is completely bipartite.

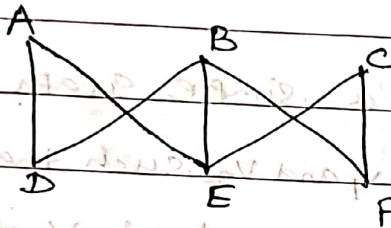
(a)



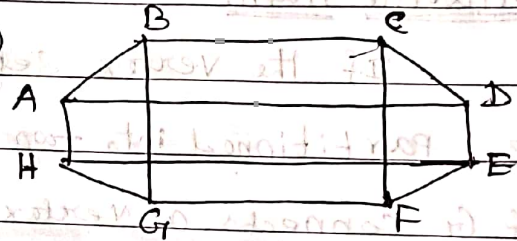
(b)



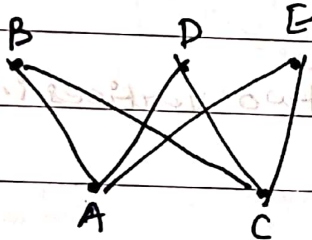
(c)



(d)

Sol:-

(b)



completely bipartite

$$V_1 = \{A, C\}$$

$$V_2 = \{B, D, E\}$$

(c)

$$V_1 = \{A, B, C\}$$

$$V_2 = \{D, E, F\}$$

B, Not a w. b. g.

(d)

$$V_1 = \{A, C, E, G\}$$

$$V_2 = \{B, D, F, H\}$$

B, Not a completely bipartite

$$(a) V_1 = \{D, E, F\}$$

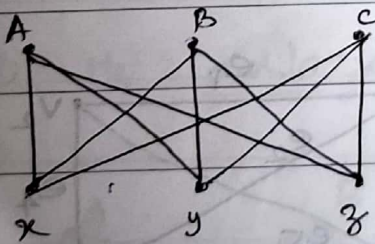
$$V_2 = \{A, B, C\}$$

Not B, Not C, B

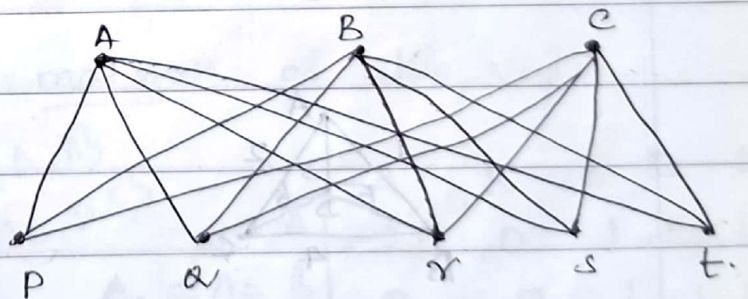
EX: Draw the complete Bipartite graphs of $K_{3,3}$, $K_{3,5}$ and $K_{2,6}$.

Sol:-

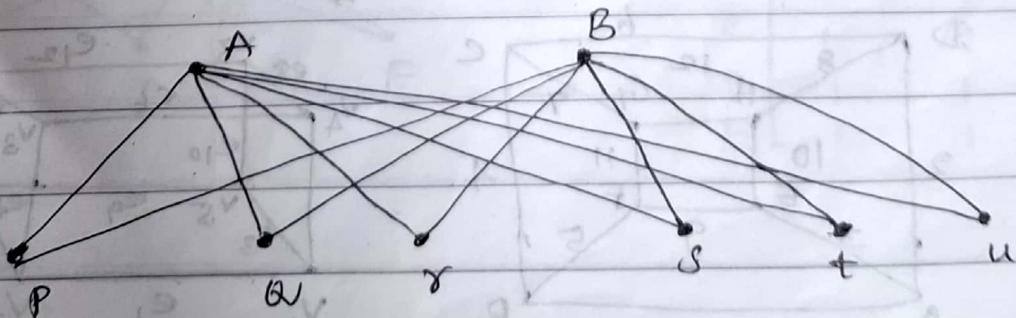
Draw $K_{3,3}$.



Draw $K_{3,5}$



Draw $K_{2,6}$



Isomorphic Graphs:

Two graphs G_1 and G_2 are said to be isomorphic to each other, if there exists a 1-1 correspondence between the vertex sets which preserves adjacency of the vertices.

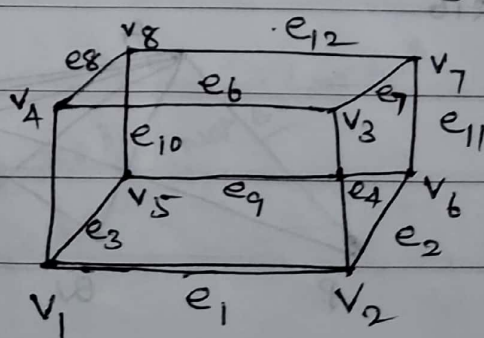
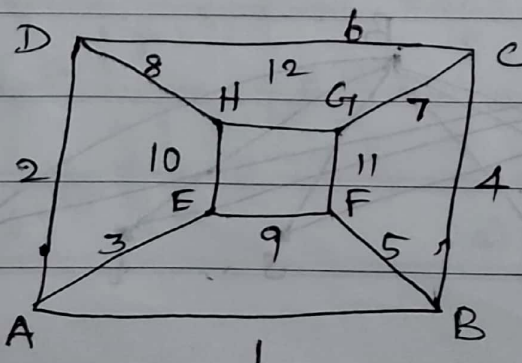
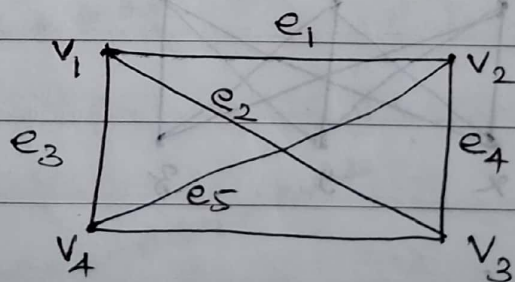
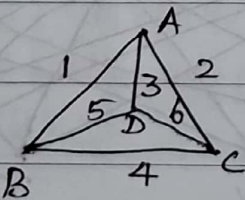
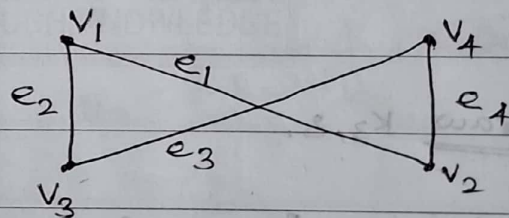
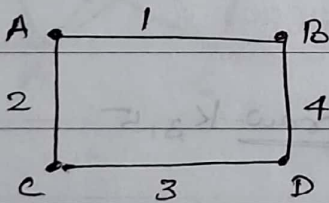
Note: We observe that isomorphic graphs have

- (1) the same number of vertices.
- (2) the same number of edges.
- (3) An equal no. of vertices with an given degree.
- (4) Equal no. of adjacencies.

This property is called an invariant with respect to isomorphic graphs. If any of these conditions is not satisfied with respect to isomorphic graphs.

In two graphs they cannot be isomorphic.

EX:



MATRIX REPRESENTATION OF GRAPHS

Adjacency matrix of a undirected graph:

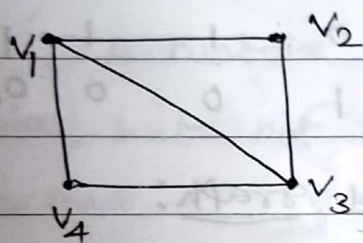
When G is a simple graph with n -vertices $v_1, v_2, \dots, v_n$
 the matrix A (or) $A_G \equiv [a_{ij}]$

where

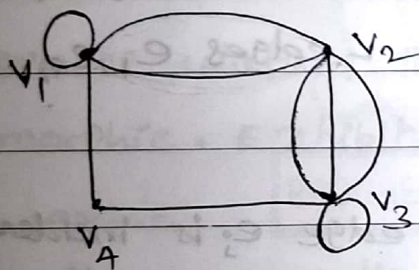
$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

is called the adjacency matrix of G .

EX:



$$A(G) = A_G = A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

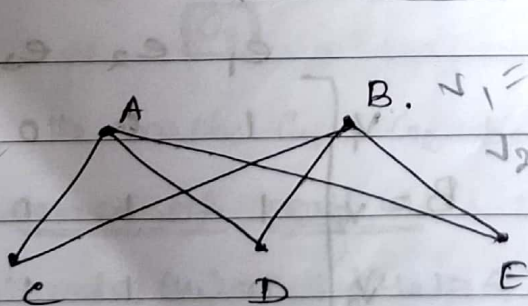


$$A_G = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

EX:

write the adjacency matrix of $K_{2,3}$.

Sol:



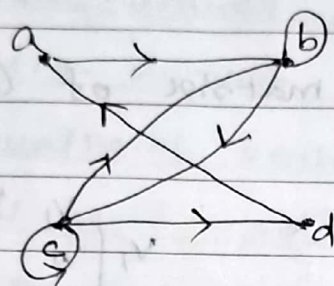
$$A_G = A = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Adjacency matrix of a directed graph:

The Adjacency matrix of a digraph G , with n -vertices is the matrix $A = [a_{ij}]_{n \times n}$ in which

$$A = [a_{ij}] = \begin{cases} 1 & ; (v_i, v_j) \text{ is a directed edge in } G \\ 0 & ; \text{ otherwise.} \end{cases}$$

Ex:



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Incidence matrix for undirected graph:

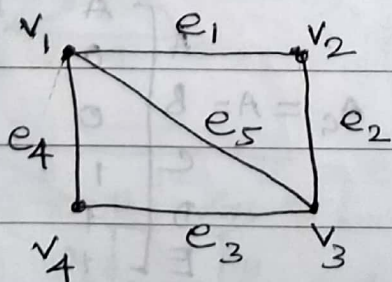
If $G = (V, E)$ is an undirected graph with n -vertices $v_1, v_2, \dots, v_n$ and m -edges $e_1, e_2, \dots, e_m$ then the $(n \times m)$ matrix $B = [b_{ij}]$

where

$$b_{ij} = \begin{cases} 1 & ; \text{ when edge } e_j \text{ is incident on } v_i \\ 0 & ; \text{ otherwise.} \end{cases}$$

is called the incident matrix of G .

Ex:



$$B = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Theorem:

Two Graphs are isomorphic, if and only if their vertices can be labeled in such a way that the corresponding adjacency matrices are equal.

Theorem:

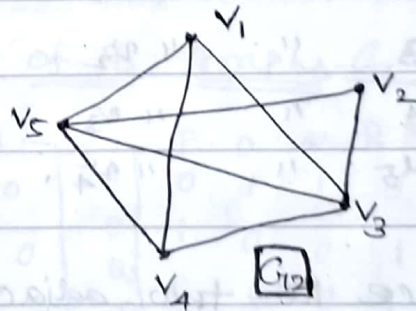
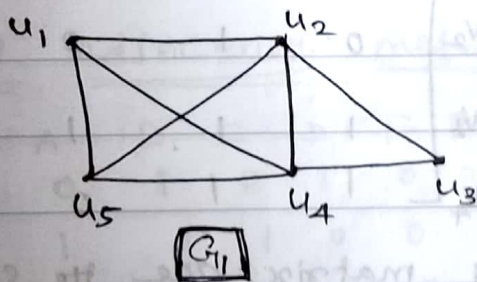
Two Labeled Graphs G_1 and G_2 with adjacency matrices A_1 and A_2 respectively are isomorphic, iff there exists a permutation matrix P such that $PA_1P^T = A_2$.

Define: PERMUTATION MATRIX:

A matrix whose rows are the rows of the unit matrix, but not necessarily in their natural order is called a permutation matrix.

=

- ① Determine whether the following pairs of graphs are isomorphic. Exhibit the isomorphism or prove that it does not exist.

Sol:

Given the graphs G_1 & G_2 both have 5 vertices and 8 edges.

To find degrees of vertices:

For G_1 : $\deg(u_1) = 3$; $d(u_2) = 4$; $d(u_3) = 2$; $d(u_4) = 4$; $d(u_5) = 3$.

For G_2 : $\deg(v_1) = 3$; $d(v_2) = 2$; $d(v_3) = 4$; $d(v_4) = 3$; $d(v_5) = 4$.

$$\sum \deg(G_1) = 3 + 4 + 2 + 4 + 3 = 16 ; \sum d(G_2) = 16$$

To find Degree Sequence of G_1 :

2, 3, 3, 4, 4. ~~3, 4, 2, 4, 3~~

To find Degree Sequence of G_2 :

2, 3, 3, 4, 4. ~~3, 2, 4, 3, 4~~

To find Adjacency matrices of the graph G_1 & G_2 :

$$G_1 = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$G_2 = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

The vertex

u_1 corresponds to v_1 , $G_2 =$

u_2 " " v_5

u_3 " " v_2

u_4 " " v_3

u_5 " " v_4

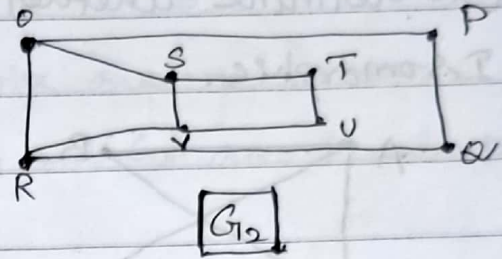
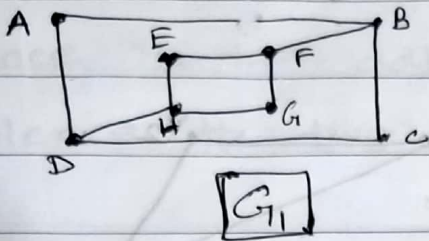
$$G_2 = \begin{matrix} & \begin{matrix} v_1 & v_5 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_5 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Since the two adjacency matrices are the same, the two graphs are isomorphic.

Note:- we assume arbitrarily that the vertex u_1 corresponds to v_1 , u_2 corresponds to v_5 ... etc. and find the adjacency matrices of the two graphs. If this choice of corresponding vertices does not lead to identical adjacency matrices, we may try another

choice of corresponding using the adjacency of vertices and degrees.

② Determine whether the graphs shown in diagram are isomorphic.



Sol:-

The graphs G_1 & G_2 both have 8-vertices and 10-edges.

Degree of vertices:

$$d(A)=2; d(B)=3; d(C)=2; d(D)=3; d(E)=2; d(F)=3; d(G)=2$$

$$d(H)=3; \quad \sum d(G_1) = 2+3+2+3+2+3+2+3 = 20.$$

$$d(P)=2; d(Q)=2; d(R)=3; d(S)=3; d(W)=3; d(T)=2; d(U)=2$$

$$d(V)=3; \quad \sum d(G_2) = 2+2+3+3+3+2+2+3 = 20$$

Degree sequence of G_1 :

$$2, 3, 2, 3, 2, 3, 2, 3$$

$$2, 2, 2, 2, 3, 3, 3, 3.$$

Degree sequence of G_2 :

$$2, 2, 2, 2, 3, 3, 3, 3.$$

To find Adjacency matrices of the graphs G_1 & G_2 :

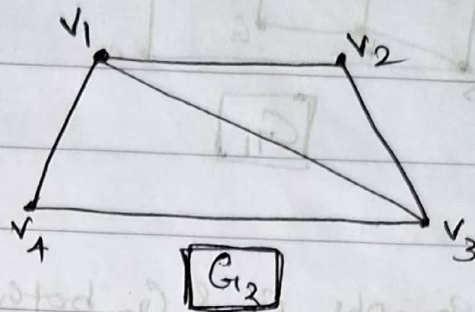
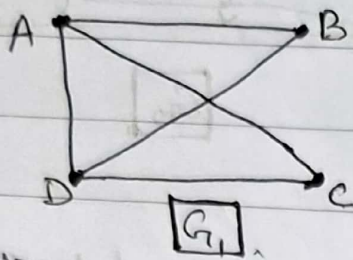
	A	B	C	D	E	F	G	H
A	0	1	0	1	0	0	0	0
B	1	0	1	0	0	1	0	0
C	0	1	0	1	0	0	0	0
D	1	0	1	0	0	0	0	1
E	0	0	0	0	0	1	0	1
F	0	1	0	0	1	0	1	0
G	0	0	0	0	0	0	1	0
H	0	0	0	0	1	1	0	1

	P	Q	R	T	S	U	V
P	0	1	1	0	0	0	0
Q	1	0	0	1	0	1	0
R	1	0	0	1	0	0	0
T	0	1	1	0	0	0	1
S	0	0	0	0	0	1	0
U	0	1	0	0	1	0	0
V	0	0	0	1	0	1	1

$\deg(a) = 2$
 since each of the vertices v_2, v_3, v_6 & v_7 has $\deg 2$ in G_1 .
 Then for a in G must correspond to either v_2, v_3, v_6 or v_7 .
 The two adjacent matrices are not equal.

Hence the two graphs G_1 and G_2 are not isomorphic.
 But v_2, v_3, v_6 & v_7 are adjacent to another vertex of degree 2, which is not the case in G_1 .

③. Determine whether the graphs shown in diagram are Isomorphic.



Sol:-

Given the graphs G_1 & G_2 both have 4-vertices & 5-edges.

Degree of vertices:

$$d(A) = 3; d(B) = 2; d(C) = 2; d(D) = 3 \quad \therefore \sum d(G_1) = 10$$

$$d(v_1) = 3; d(v_2) = 2; d(v_3) = 2; d(v_4) = 3 \quad \therefore \sum d(G_2) = 10$$

To find Degree Sequence of G_1 & G_2 :

2, 2, 3, 3

& 2, 2, 3, 3.

To find adjacency matrices of the Graphs G_1 & G_2 :

$$G_1 = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$G_2 = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$G_2 = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_4 & v_3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_4 \\ v_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

The vertex A corresponds to v_1

B " " " v_2

C " " " v_4

D " " " v_3

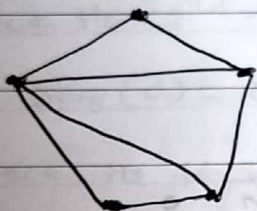
Since the two adjacency matrix are the same.

Hence the two graphs G_1 and G_2 are isomorphic.

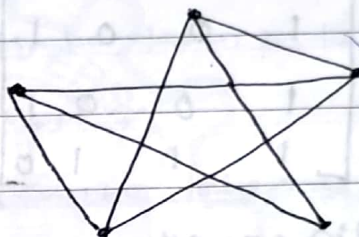
$\Rightarrow \alpha =$

④ Examine whether the following pairs of graphs are isomorphic. If not isomorphic, give the reasons.

(1)

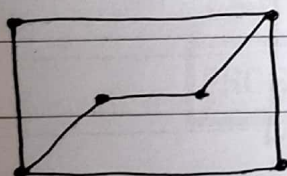


G_1

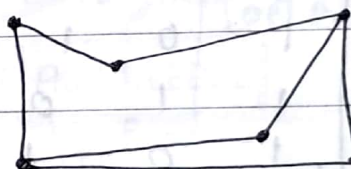


G_2

(2)

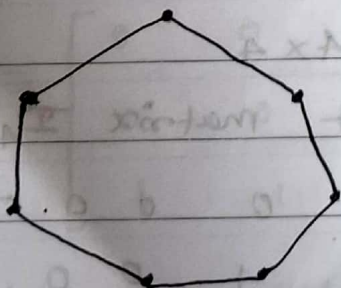


G_1

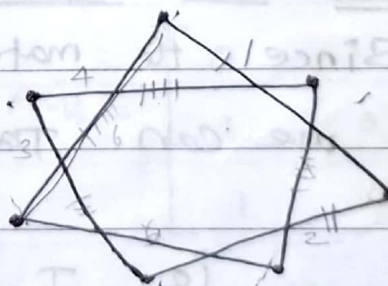


G_2

(3)

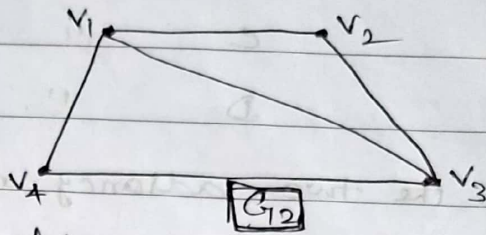
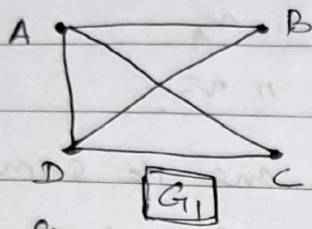


G_1



G_2

① Establish isomorphism between G_1 and G_2 in given below



to find a permutation matrix P .

Sol:-

The adjacency matrices A_{G_1} is

$$A_{G_1} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The adjacency matrices A_{G_2} is

$$A_{G_2} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

The matrices A_{G_1} & A_{G_2} are not the same.

To find ^{PERMUTATION} ~~matrix~~ matrix P :

Since the matrix is 4×4

Now we can take unit matrix I_4 .

$$\text{i.e., } I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagram - G_1 :

A	B	C	D
3	2	2	3

Diagram - G_2 :

V_1	V_2	V_3	V_4
3	2	3	2

Using degree of vertex:

$$\textcircled{1} \text{ Deg}(A) = 3 \quad \& \quad \text{Deg}(V_1) = 3.$$

Hence the 1st row of I_A can be taken as the first row of P .

$$\textcircled{2} \text{ Deg}(D) = 3 \quad \& \quad \text{Deg}(V_3) = 3.$$

Hence the 4th row of I_A can be taken as the third row of P .

$$\textcircled{3} \text{ Deg}(B) = 2 \quad \& \quad \text{Deg}(V_2) = 2$$

Hence the 2nd row of I_A can be taken as the second row of P .

$$\textcircled{4} \text{ Deg}(C) = 2 \quad \& \quad \text{Deg}(V_4) = 2$$

Hence the third row of I_A can be taken as the fourth row of P .

$$\therefore P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad [\text{OR}] \quad P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Hence the two graphs G_1 and G_2 are isomorphic.

H.W

$$AG_1 \equiv \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$AG_2 \equiv \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

To find Permutation matrix.

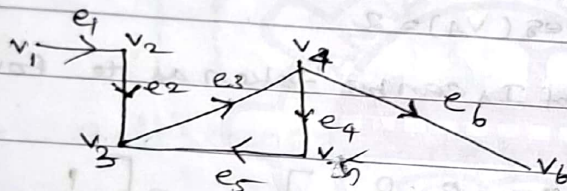
Definitions:Walk:

A walk in an undirected graph $G=(V,E)$ is a finite non-empty sequence $w=v_0 e_1 v_1 e_2 v_2 e_3 \dots v_{k-1} e_k v_k$ consisting of vertices and edges alternatively such that e_i connects v_{i-1} and v_i .

The vertices $v_0 \rightarrow$ origin of walk

$v_k \rightarrow$ terminus of walk.

and other vertices are called internal vertices of the walk.

Ex:

$v_1 \xrightarrow{e_1} v_2 \xrightarrow{e_2} v_3 \xrightarrow{e_3} v_4 \xrightarrow{e_4} v_5 \xrightarrow{e_5} v_3 \xrightarrow{e_3} v_4 \xrightarrow{e_6} v_6$ is called walk.

Path:

A path in a graph is finite alternating sequence of vertices and edges beginning and ending with vertices

Ex: (from above diagram)

$v_1 \xrightarrow{e_1} v_2 \xrightarrow{e_2} v_3 \xrightarrow{e_3} v_4 \xrightarrow{e_6} v_6$ is called a path.

Simple Path:

If the edges in a path are distinct it is called a simple path.

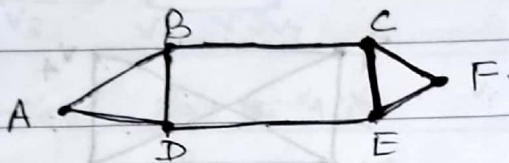
cycle (or) circuit:-

A path with originates and ends in the same vertex is called a cycle (or) circuit.

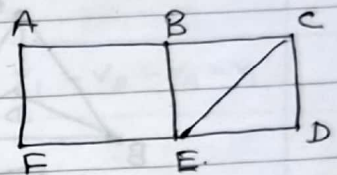
Ex (from above diagram)

$V_2 \xrightarrow{e_3} V_4 \xrightarrow{e_4} V_5 \xrightarrow{e_5} V_2$ is called a circuit.

① Find all the simple paths from A to F and all the circuits in the graph



H.W



Sol:-

The simple paths from A to F are the following

- (1) A - B - C - F (2) A - D - E - F (3) A - B - D - E - F
 (4) A - D - B - C - F (5) A - B - C - E - F (6) A - D - E - C - F
 (7) A - B - D - E - C - F (8) A - D - B - C - E - F.

The circuits in the graph are the following

- (1) A - B - D - A (2) C - F - E - C (3) B - C - E - D - B
 (4) A - B - C - E - D - A (5) B - C - F - E - D - B (6) A - B - C - F - E - D - A.

② Find all the connected subgraph obtained from the graph given in the following diagram by deleting each vertex list out the simple paths from A to F in each subgraph.

Sol:- ① There is no simple path from A to F.

② Simple path A - F.

③ Simple path A - F
 A - B - E - F.

④ Simple path A - F
 A - B - E - F
 A - B - C - E - F

⑤ Simple path A - F

⑥ There is no simple path.

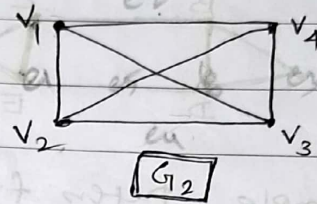
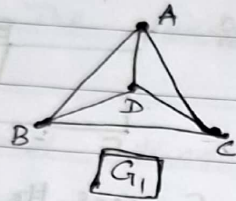
Circuits and Isomorphism: [we see Page: 18]

① vertices ② edges ③ degree. ④ adjacency.

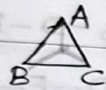
The 4-conditions are satisfied. ^{is} isomorphic. (Also find circuit).

The 4-conditions are not satisfied is not is not isomorphic.

① Using circuits, examine whether the following pairs of graphs G_1 and G_2 given below are isomorphic or not.



Sol:-



Given the graphs G_1 & G_2 have 4-vertices and 6-edges.

Given the graphs G_2 have 4-vertices and 6-edges.

To find degree of vertices:

For G_1 : $d(A)=3$; $d(B)=3$; $d(C)=3$; $d(D)=3$. [$\sum d(G_1)=12$]

For G_2 : $d(v_1)=3$; $d(v_2)=3$; $d(v_3)=3$; $d(v_4)=3$. [$\sum d(G_2)=12$]

To find Adjacency matrices of the graphs G_1 & G_2 :

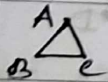
$$G_1 = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$G_2 = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

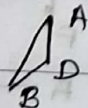
Since the two adjacency matrix are the same, the two graphs are isomorphic.

To find circuits of length:

The circuits of length 3 exist in G_1 .



A-B-C-A



A-D-B-A

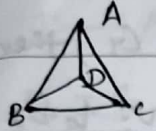


A-C-D-A

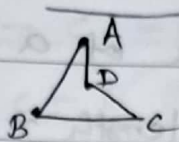


B-C-D-B.

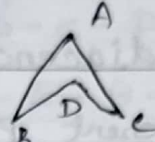
the circuits of length 4 in G_1 :



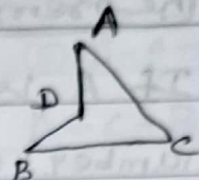
$A-B-C-D-A$



$A-B-C-D-A$

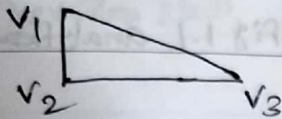


$A-B-D-C-A$

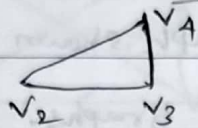


$A-D-B-C-A$

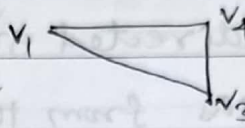
the circuits of length 3 in G_2 :



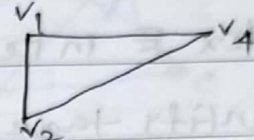
$v_1-v_2-v_3-v_1$



$v_2-v_3-v_4-v_2$

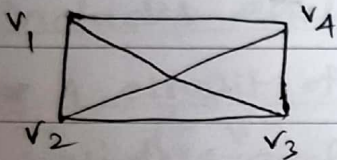


$v_1-v_3-v_4-v_1$



$v_1-v_2-v_4-v_1$

the circuits of length 4 in G_2 :



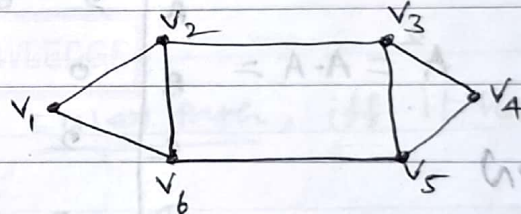
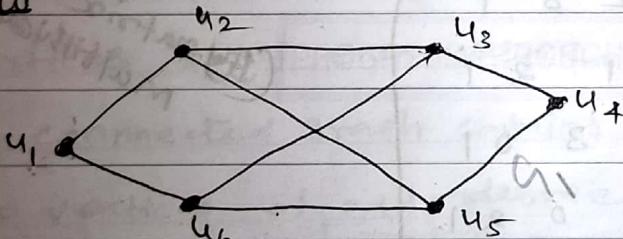
$v_1-v_2-v_4-v_3-v_1$ & $v_1-v_4-v_2-v_3-v_1$

$v_1-v_2-v_3-v_4-v_1$

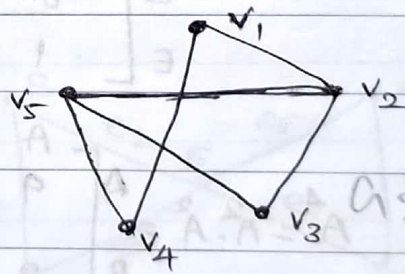
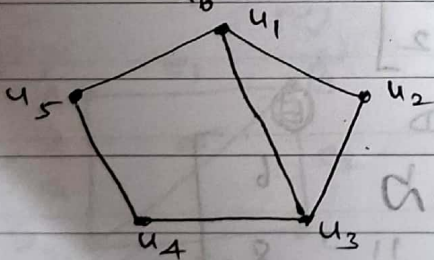
Hence the two graphs are isomorphic.

H.W

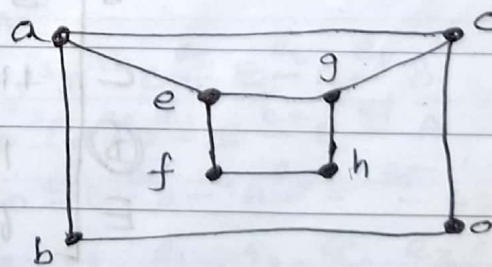
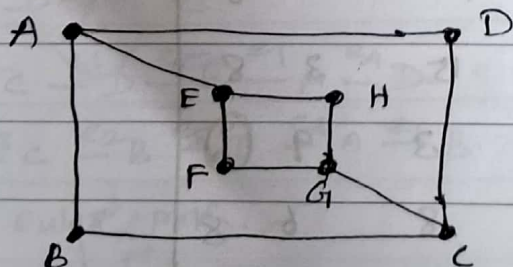
(1)



(2)



(3)



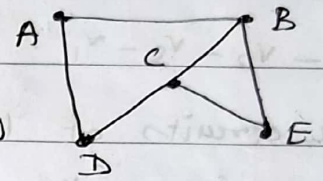
Theorem:

If A is the adjacency matrix of a graph G , then the number of different paths of length r from v_i to v_j is equal to the (i, j) th entry of A^r .

① Find the number of paths of length 4 from the vertex D to the vertex E in the undirected graph shown in Fig 1.1. analytically. Identify those paths from the graphs.

Sol:-

The adjacency matrix of the given graph is



$$A = \begin{bmatrix} & A & B & C & D & E \\ A & 0 & 1 & 0 & 1 & 0 \\ B & 1 & 0 & 1 & 0 & 1 \\ C & 0 & 1 & 0 & 1 & 1 \\ D & 1 & 0 & 1 & 0 & 0 \\ E & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} & A & B & C & D & E \\ A & 2 & 0 & 2 & 0 & 1 \\ B & 0 & 3 & 1 & 2 & 1 \\ C & 2 & 1 & 3 & 0 & 1 \\ D & 0 & 2 & 0 & 2 & 1 \\ E & 1 & 1 & 1 & 1 & 2 \end{bmatrix}$$

(Matrix multiplication)

$$A^4 = A^2 \cdot A^2 = \begin{bmatrix} & A & B & C & D & E \\ A & 9 & 3 & 11 & 1 & 6 \\ B & 3 & 15 & 7 & 11 & 8 \\ C & 11 & 7 & 15 & 3 & 8 \\ D & 1 & 11 & 3 & 9 & 6 \\ E & 6 & 8 & 8 & 6 & 8 \end{bmatrix}$$

Hence there are 6 paths each of length 4 from D to E.

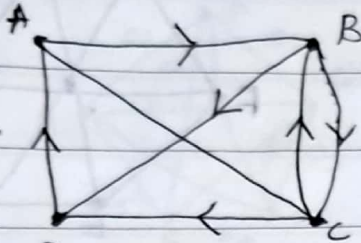
- ① $D-A-D-C-E$ ② $D-C-D-C-E$ ③ $D-A-B-C-E$
 ④ $D-C-E-C-E$ ⑤ $D-C-E-B-E$ ⑥ $D-C-B-C-E$

— X —

H.W

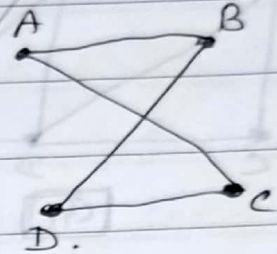
①

Reverse B to D.



②

Reverse A to D.



EULERIAN GRAPHS:

* A path of graph G is called an Eulerian Path if it includes each edge of G exactly once.

* A circuit of a graph G is called an Eulerian circuit, if it includes each edge of G exactly once.

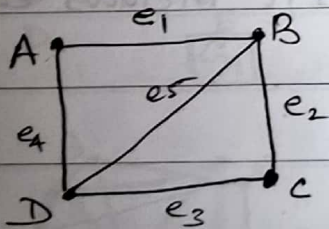
* A graph containing an Eulerian circuit is called an Eulerian graph.

Theorem:

① A connected graph contains an Euler circuit, iff each of its vertices is of even degree.

② A connected graph contains an Euler path, iff it has exactly two vertices of odd degree.

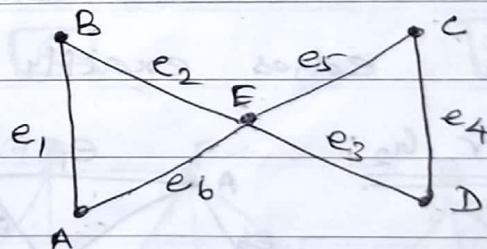
EX:-



$B \xrightarrow{e_2} C \xrightarrow{e_3} D \xrightarrow{e_4} B \xrightarrow{e_1} A \xrightarrow{e_5} D$

$D \xrightarrow{e_3} C \xrightarrow{e_2} B \xrightarrow{e_5} D \xrightarrow{e_4} A \xrightarrow{e_1} B$

is Euler path.



$B \xrightarrow{e_2} E \xrightarrow{e_3} D \xrightarrow{e_4} C \xrightarrow{e_5} E \xrightarrow{e_6} A \xrightarrow{e_1} B$

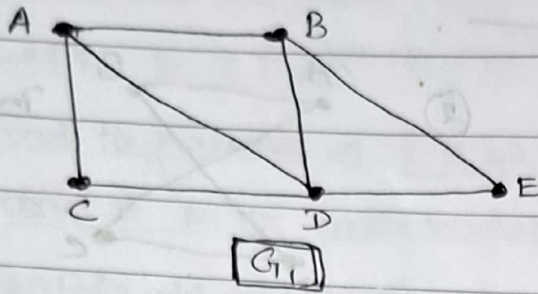
$A \xrightarrow{e_6} E \xrightarrow{e_5} C \xrightarrow{e_4} D \xrightarrow{e_3} E \xrightarrow{e_2} B \xrightarrow{e_1} A$

$C \xrightarrow{e_4} D \xrightarrow{e_3} E \xrightarrow{e_2} B \xrightarrow{e_1} A \xrightarrow{e_6} E \xrightarrow{e_5} C$

$D \xrightarrow{e_3} E \xrightarrow{e_2} B \xrightarrow{e_1} A \xrightarrow{e_6} E \xrightarrow{e_5} C \xrightarrow{e_4} D$

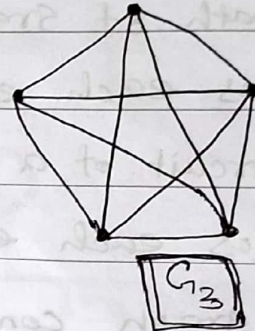
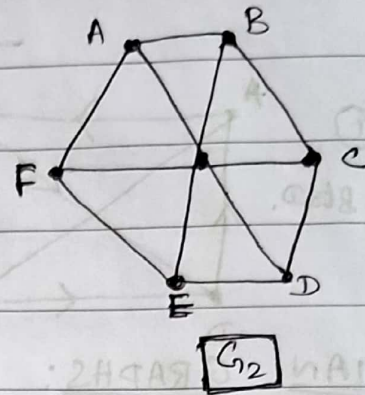
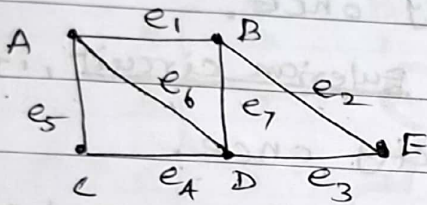
is Euler circuit.

- ① Find an Euler path or an Euler circuit, if it exists. In each of the three graphs shown is given below. If it does not exist, explain why?



Sol:-

For G_1



There are only two vertices A and B of degree 3 and other vertices are even degree.

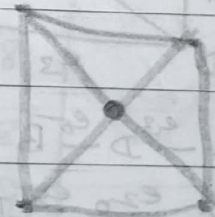
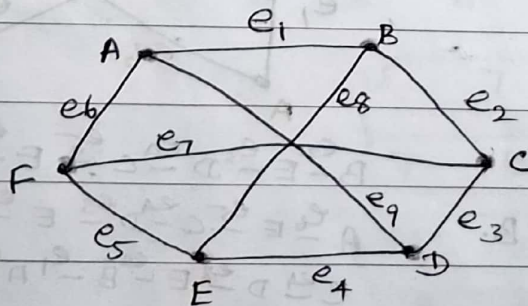
By theorem, Hence there exists an Euler path b/w A & B. The path is " " " b/w B & A.

A $\xrightarrow{e_1}$ B $\xrightarrow{e_2}$ E $\xrightarrow{e_3}$ D $\xrightarrow{e_6}$ A $\xrightarrow{e_5}$ C $\xrightarrow{e_4}$ D $\xrightarrow{e_7}$ B.

B $\xrightarrow{e_1}$ A $\xrightarrow{e_6}$ C $\xrightarrow{e_5}$ D $\xrightarrow{e_3}$ E $\xrightarrow{e_2}$ B $\xrightarrow{e_4}$ D $\xrightarrow{e_7}$ A.

This is an Eulerian Path, as it includes each of the edges exactly once.

For G_2 :

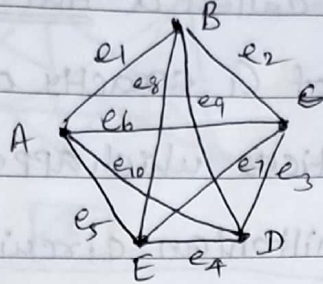


There are 6 vertices of odd degree.

By Theorem

This is an Eulerian Path.

For G_3 :



All the vertices are of even degree. Hence there exists an Euler circuit.

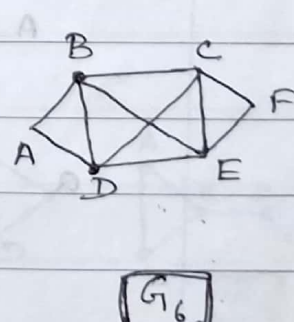
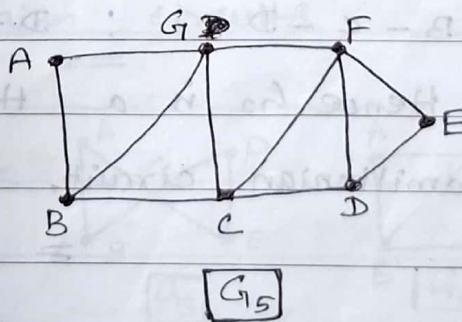
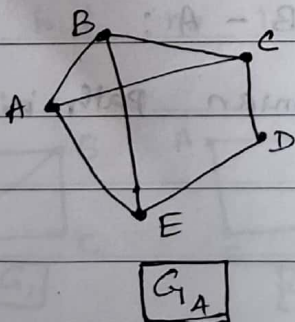
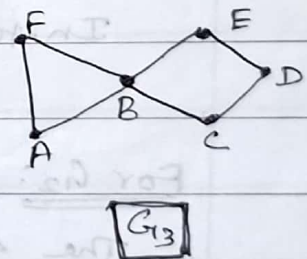
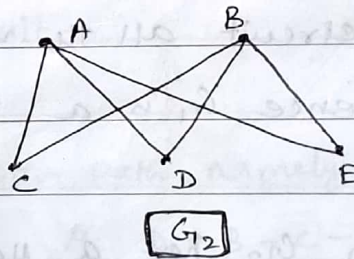
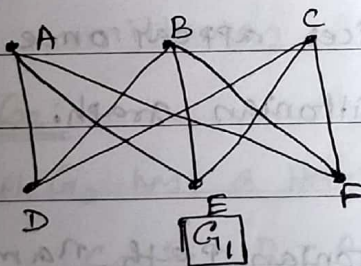
The Eulerian circuit is

$A \xrightarrow{e_1} B \xrightarrow{e_2} C \xrightarrow{e_3} D \xrightarrow{e_4} E \xrightarrow{e_5} A \xrightarrow{e_6} C \xrightarrow{e_7} E \xrightarrow{e_8} B \xrightarrow{e_9} D \xrightarrow{e_{10}} A \xrightarrow{e_1} B$
 $B \xrightarrow{e_2} C \xrightarrow{e_3} D \xrightarrow{e_4} E \xrightarrow{e_5} A \xrightarrow{e_6} C \xrightarrow{e_7} E \xrightarrow{e_8} B \xrightarrow{e_9} D \xrightarrow{e_{10}} A \xrightarrow{e_1} B$

Since it includes each of the 10 edges exactly once.

H.W

① Find an Euler path or an Euler circuit, if it exists in each of the six graphs shown in given below. If it does not exist, explain why?

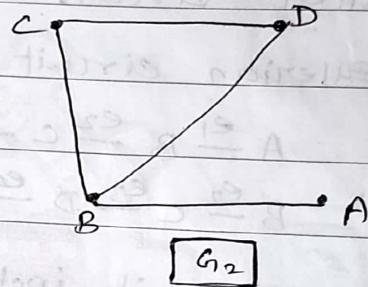
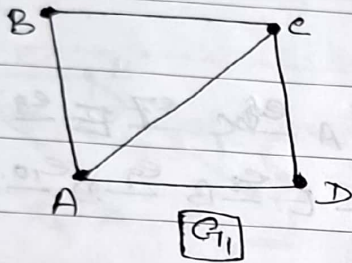




HAMILTONIAN GRAPHS

- * A path of a graph G is called a Hamiltonian Path, if it includes each vertex of G exactly once.
- * A circuit of a graph G is called a Hamiltonian circuit, if it includes each vertex of G exactly once, except the starting and end vertices which appear twice.
- * A graph containing a Hamiltonian circuit is called a Hamiltonian graph.

EX:



For G_1 :

The graph G_1 has a Hamiltonian circuit, namely

$A - B - C - D - A$; $B - C - D - A - B$;

$C - D - A - B - C$; $D - A - B - C - D$;

In this circuit all the vertices appear once.

Hence G_1 is a Hamiltonian graph.

For G_2 :

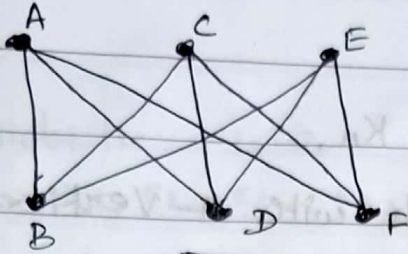
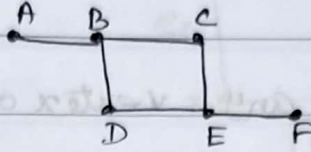
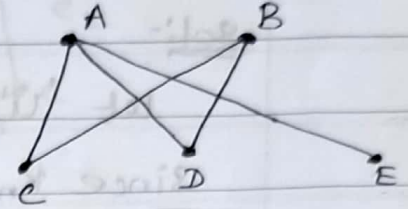
The graph G_2 has a Hamiltonian path, namely

$A - B - C - D$; $D - C - B - A$;

Hence G_2 is a Hamiltonian path, is not a Hamiltonian circuit.

$\equiv$

Find a Hamiltonian path or a Hamiltonian circuit, if it exists in each of the three graphs shown given below. If it does not exist, explain why?

 G_1  G_2  G_3

Sol:-

For G_1 :

The graph G_1 has a Hamiltonian circuit, namely

$A-B-C-D-E-F-A$; $A-B-C-F-E-D-A$

$A-B-E-D-C-F-A$; $A-B-E-F-C-D-A$

$A-D-C-B-E-F-A$; $A-D-E-B-C-F-A$.

In this circuit all the vertices appear once.

Hence G_1 is a Hamiltonian graph.

For G_2 :

The graph G_2 contains neither a Hamiltonian path nor a Hamiltonian circuit.

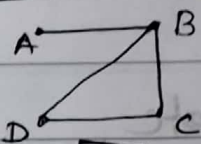
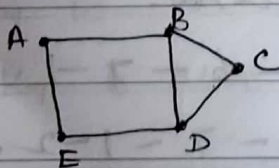
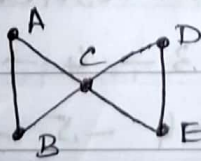
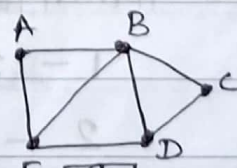
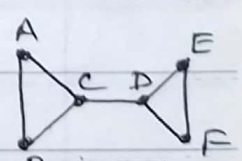
For G_3 :

The graph G_3 has a Hamiltonian path, namely

$C-B-D-A-E$ & $D-B-C-A-E$.

but no Hamiltonian circuits.

H.W:-

 G_1  G_2  G_3  G_4  G_5

① Show that complete graph on n -vertices K_n with $n \geq 3$ has Hamilton cycle. Obtain all the two edge disjoint Hamilton cycles in K_7 .

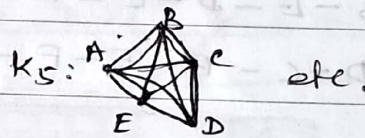
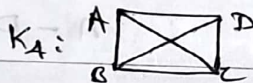
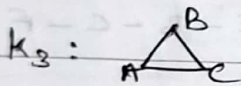
Sol:-

Let 'u' be any vertex of K_n .

Since K_n is a complete graph with n -vertices, any two vertices are joined. So, we start with 'u' and visit vertices in any order exactly once and come back to 'u'.

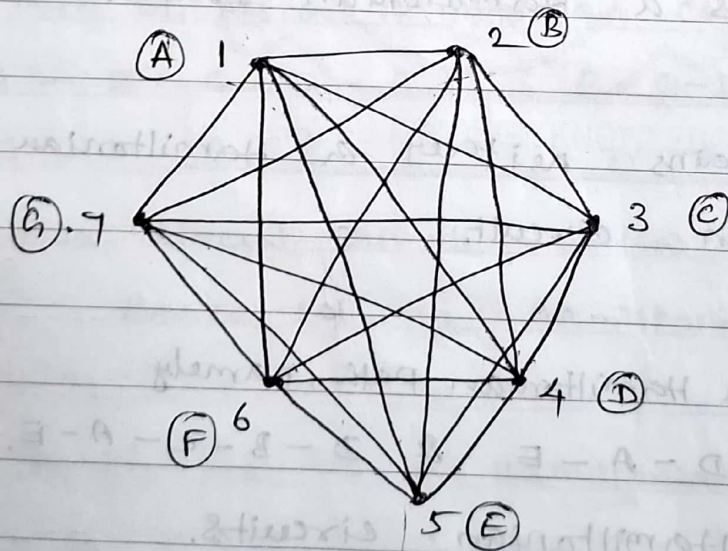
Hence, there is a Hamiltonian cycle in K_n and thus K_n is Hamiltonian.

Ex: K_n , $n \geq 3$.



etc.

The diagram of K_7 :



The Hamiltonian cycles are

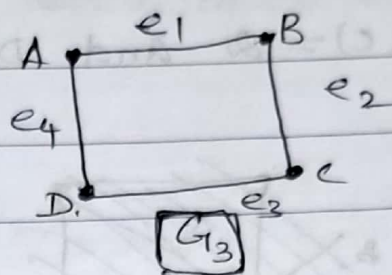
1 - 2 - 3 - 4 - 5 - 6 - 7 - 1

2 - 3 - 4 - 5 - 6 - 7 - 1 - 2 ... etc

Draw a graph that is both Eulerian and Hamiltonian.

Sol:-

Consider the graph



Consider the cycle

$$A \xrightarrow{e_1} B \xrightarrow{e_2} C \xrightarrow{e_3} D \xrightarrow{e_4} A$$

$$B \xrightarrow{e_2} C \xrightarrow{e_3} D \xrightarrow{e_4} A \xrightarrow{e_1} B ; C \xrightarrow{e_3} D \xrightarrow{e_4} A \xrightarrow{e_1} B \xrightarrow{e_2} C ; D \xrightarrow{e_4} A \xrightarrow{e_1} B \xrightarrow{e_2} C \xrightarrow{e_3} D.$$

Since the cycle contains all the edges, G_3 is Eulerian.
- exactly once.

Since the cycle contains all the vertices exactly once

THE GRAPH G_3 is Hamiltonian.

$\therefore G_3$ is an example of both Eulerian and Hamiltonian.

— x —.

③ Show that K_7 has Hamiltonian graph. How many edge disjoint Hamiltonian cycles are there in K_7 ? List all the edge-disjoint Hamiltonian cycles. Is it Eulerian graph?

Sol:-

The graph of K_7 . [we see page 38]

K_7 has two edge disjoint Hamiltonian cycles.

The edge disjoint Hamiltonian cycles are

$$1-2-3-4-5-6-7-1 \text{ and}$$

$$1-3-6-2-4-7-5-1.$$

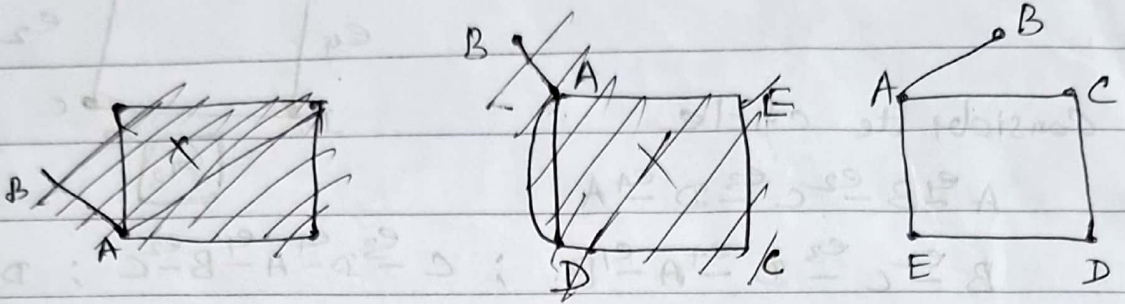
$\therefore K_7$ is Hamiltonian graph.

K_7 is also Eulerian graph.

— x —.

- ④ Draw the graph with 5 vertices A, B, C, D and E such that the $\deg(A) = 3$, B is an odd vertex, $\deg(C) = 2$ and D and E are adjacent.

Sol:-

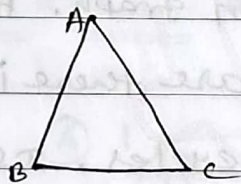


- ⑤ Show that the complete graph with n -vertices K_n has a Hamiltonian circuit whenever $n \geq 3$.

Sol:-

[We see page : 38]

The graph of K_3 .



The Hamiltonian cycles are

$A-B-C-A$; $B-C-A-B$; $C-A-B-C$

Hence K_3 is Hamiltonian cycles.

- ⑥ Check whether the two graphs given in Fig: Q13 are isomorphic or not.

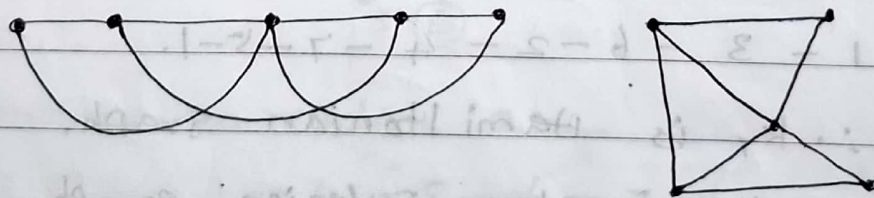
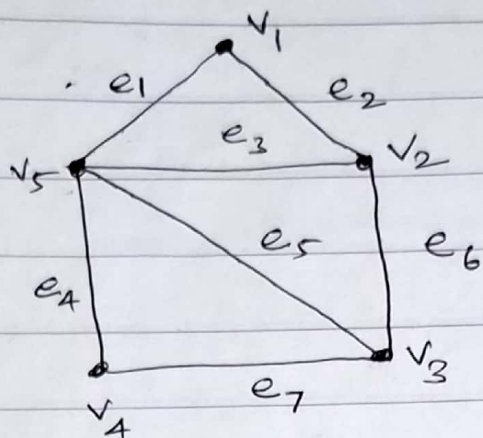


Fig : Q13.

⑦ Obtain the adjacency matrix of the graph given below:



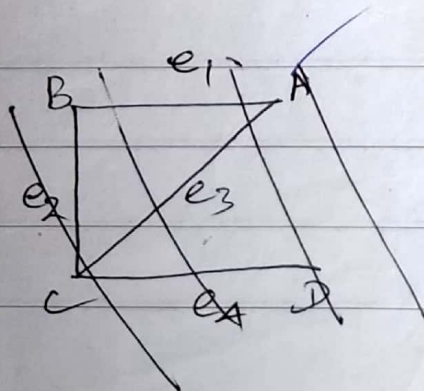
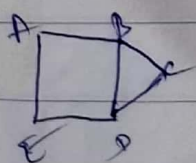
Sol:-

The adjacency matrix is

$$AG \equiv \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

⑧ Give an example of a non-Eulerian graph which is Hamiltonian.

Sol:-



~~$A-B-C-D$ is Hamiltonian path.
(vertex exactly once).~~

~~$A \xrightarrow{e_1} B \xrightarrow{e_2} C \xrightarrow{e_4} D$ is non-Eulerian graph.~~

~~Since the e_3 is missing.~~

UNIT-5

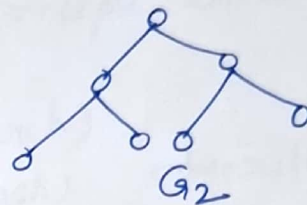
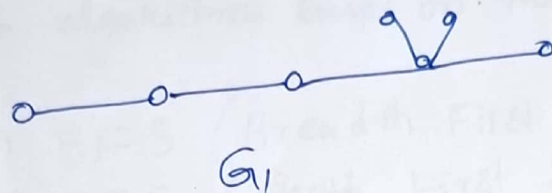
Graph Theory

①

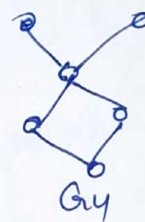
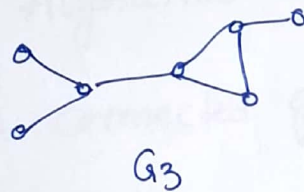
Trees: Trees form one of most widely used subclasses of graphs.

Def: A Tree is connected, undirected simple acyclic graph. In other words, a tree is a simple graph G such that there is a unique simple undirected path between each pair of vertices of G .

EX:

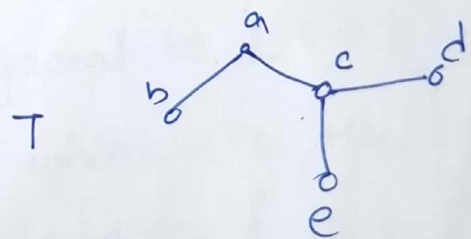
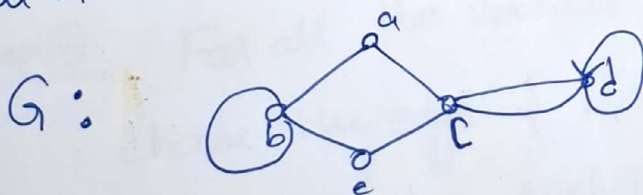


G_1 and G_2 are trees



G_3 and G_4 are not trees since they have cycles.

Spanning trees A tree T is a spanning tree of a graph G if T is a subgraph of G that contains all of the vertices of G .



a spanning tree of G

Note: A spanning tree that is directed tree is called a directed spanning tree of G .

BFS (Breadth First Spanning Trees) & DFS (Depth First Spanning Trees) ②

An algorithm for finding spanning trees by removing edges from simple cycles is inefficient, since it requires that simple cycles to be identified. Instead of constructing spanning trees, by removing edges, spanning trees can be built by successively adding edges.

Two algorithms based on this principle will be presented here

- i) BFS (Breadth First Search)
- ii) DFS (Depth First Search) algorithms.

(i) DFS Algorithm

Input: A connected graph G with vertices $v_1, v_2, v_3, \dots, v_m$.

Output: A spanning tree T for G .

Step ① Let v_1 be initial vertex. Set $L = v_1$ (L stands for the vertex last visited).

Step ② For all the vertices adjacent to L , choose the edge $\{L, v_k\}$ where k is the minimum index such that adding $\{L, v_k\}$ to T does not create a cycle.

③ If no such edge exists, go to step 3.
otherwise add edge $\{L, v_k\}$ to T and set $L = v_k$
repeat the step 2 at the new value for L .

Step ③: If x is a parent of L in T set
 $L = x$ and apply step 2 at the new value of L .
Otherhand, L has no parent in T (so that $L = v_1$)
the DFS terminates and T is a spanning tree of G .

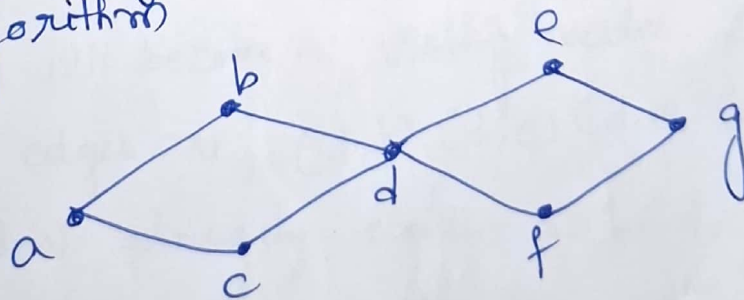
Note: → Choose a vertex as the root of the spanning tree.
arbitrarily

→ Form a path starting at this vertex by
Successively adding the edges where each
new edge is incident with the last vertex in
the path and the vertex not already in the path.

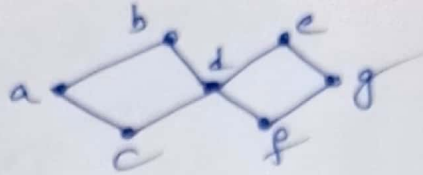
→ Continue this process by adding edges to this
path as long as possible without producing any
cycles.

→ If the path goes through all the vertices in
the graph, then the tree is a spanning tree.

Q) Find a spanning tree of the graph by using
DFS algorithm



80] Step 1 Given graph



(4)

Step 2 Given connected graph consisting 7 nodes namely a, b, c, d, e, f and g.

Step 3 choose a visiting vertex 'a'.

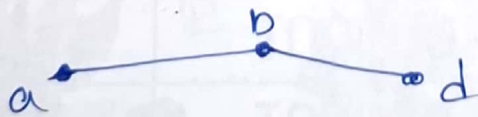
Step 4 By successively adding the edges incident with the vertices not already in the path as long as possible to the path.

i.e., the adjacent edges are (a, b) (a, c)

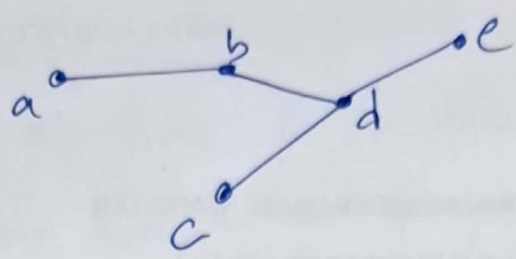
The edge (a, b) does not consist in the previous path, then we have to add the order pair edge (a, b) to the visiting vertex 'a'.



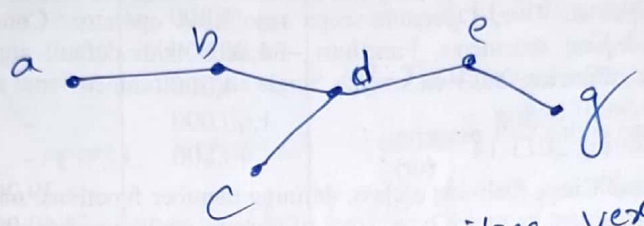
Step 5: Now the vertex 'b' will become a visiting vertex. The adjacent edge is (b, d). It is not containing in previous path. so adding the edge (b, d) to the vertex 'b'



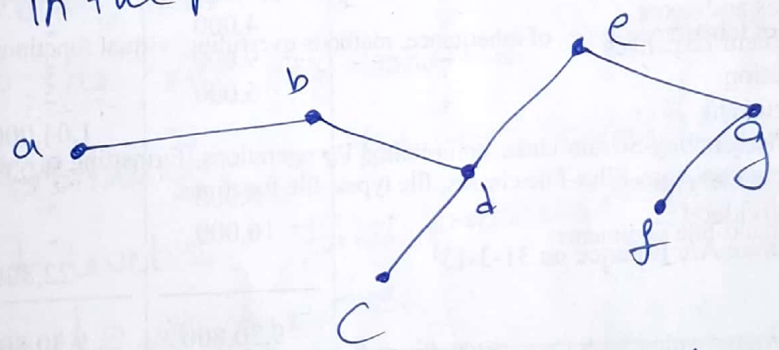
Step 6 Now d will become a visiting vertex. Its adjacent edges are (d, b) (d, c) (d, e) (d, f). In these (d, b) already exist so (d, c) (d, e) (d, f) adding (d, c) and (d, e) to the vertex d



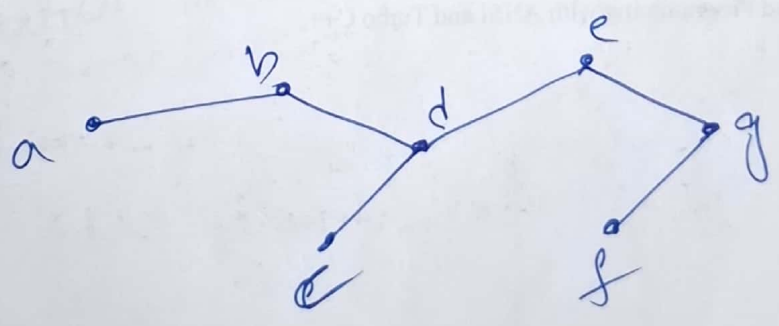
Step 7 Now e is the visiting vertex.
 It has adjacent edges (e,d) (e,g) . (e,d) already existing so we add the edge (e,g)



Step 8 : Now g is the visiting vertex. It has the adjacent edge (g,e) (g,f) . (g,e) already exists in the path. So add (g,f) to the vertex g.



Therefore Required spanning tree is



BFS Algorithm

6

The idea of BFS is to visit all vertices sequentially on a given level before going on to the next level.

Procedure: → Arbitrarily choose a root from the vertices of a graph

→ Then add all edges incident to this vertex, such that the addition of edges does not produce any cycle.

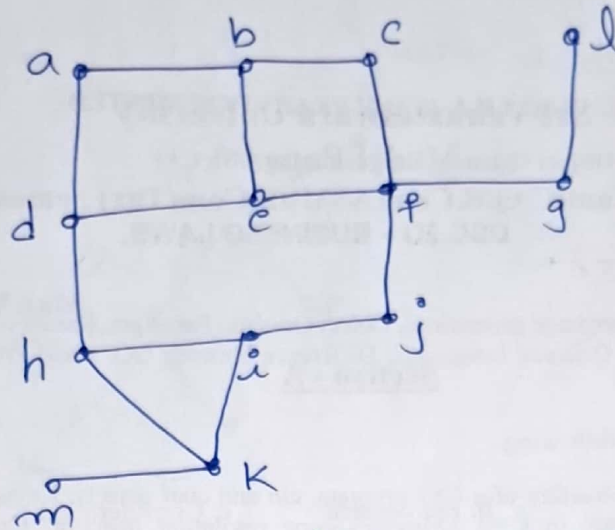
→ The new vertices added at this stage become the vertices at Level 1. In the spanning tree, arbitrarily order them.

→ Next for each vertex at Level 1, visited in the order, add each edge incident to this vertex to the tree as long as it does not produce cycle.

→ Arbitrarily order the children of each vertex at level 1. This produces the vertices at level 2 in the tree.

→ Follow the same procedure until all the vertices in the tree have been added.

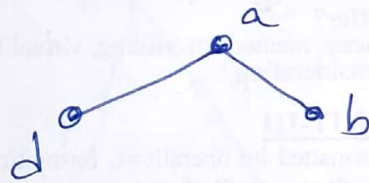
→ Use BFS to find spanning tree for the graph. ⑦



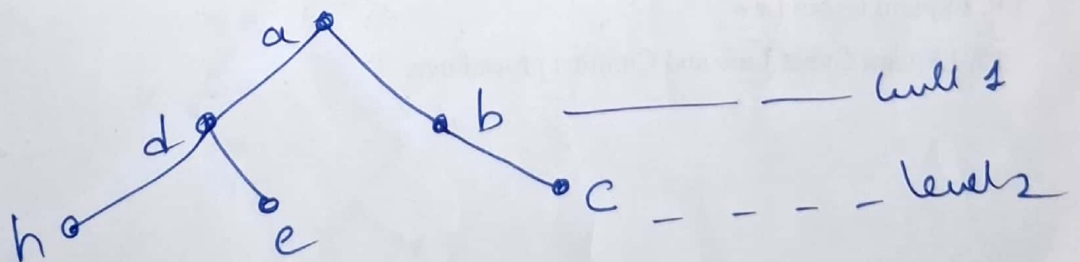
Sol: ① choose the vertex 'a' as the root

a

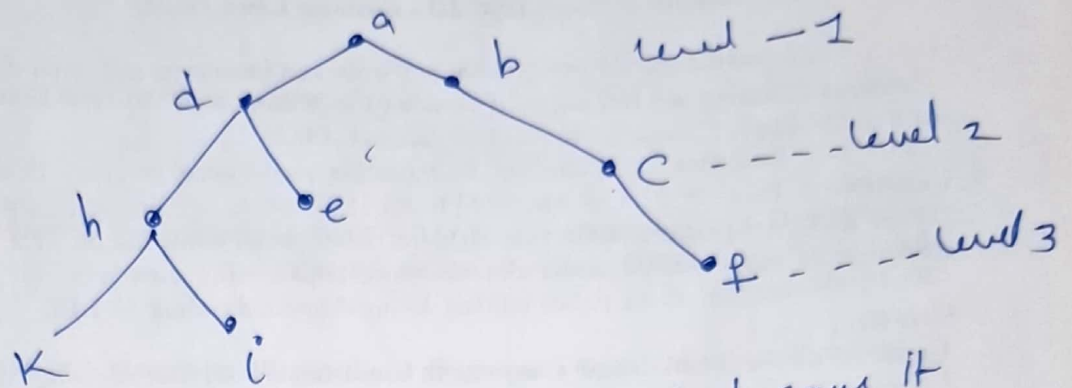
② Then add all edges incident with all vertices adjacent to a, so that edges $\{a, b\}, \{a, d\}$ are added. The vertices b, d are at Level 1 in the tree



③ Next, add the edges from these vertices at level 1 to adjacent vertices not already in the tree. Hence the edges $\{b, c\}, \{d, h\}, \{d, e\}$ are added. Now c, h, e are at level 2.

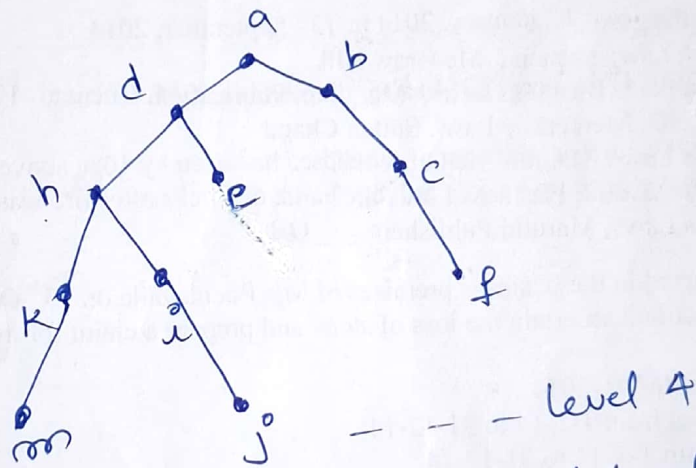


- ⑧ Next add the edges from these vertices to adjacent vertices not already in the graph. Hence the edges $\{c, f\}$, $\{b, i\}$, $\{b, k\}$ are added:

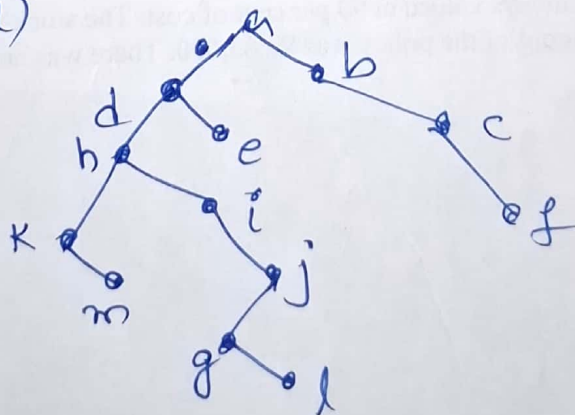


($\{c, i\}$ & $\{e, f\}$ should not be added because it forms a cycle)

- ⑤ Add $\{i, j\}$ $\{k, m\}$, now it is.



- ⑥ Add (j, g) , then g will be at level 5
(g, l)

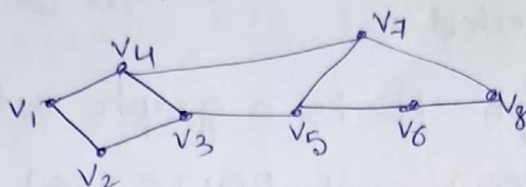


Graph Theory

①

path: In a non-directed graph G a sequence 'p' of zero or more edges of the form $\{v_0, v_1\}, \{v_1, v_2\}, \dots, \{v_{n-1}, v_n\}$ (or) $v_0 - v_1 - v_2 - \dots - v_n$ is called a path from v_0 to v_n , here v_0 is the initial vertex and v_n is the final vertex of the path P .

Ex:



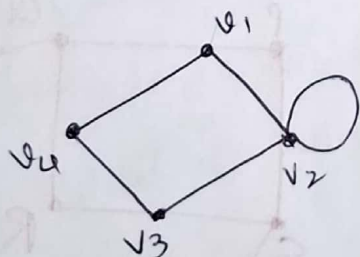
path $v_1 - v_4 - v_3 - v_5 - v_6 - v_8 - v_7 - v_4$

- Note:** 1) The no. of edges appearing in the sequence of the path is called length.
2) If the length of the path is zero, then the path P have no edges.

Circuit: A path of length ≥ 1 with no repeated edges and whose end vertices are same is called a circuit.

Note: A circuit may have repeated vertices other than the end vertices.

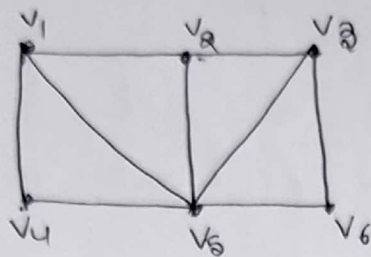
cycle: A cycle is a circuit with no other repeated vertices except its end vertices.



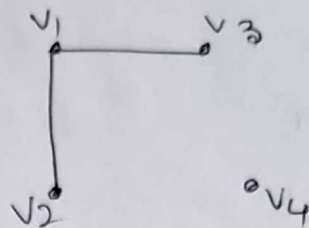
$\{v_2, v_2\}$ - cycle of length 1

$\{v_4, v_3\}, \{v_3, v_2\}, \{v_2, v_1\}$ is a cycle of length 3

Connected graph: A graph G is said to be connected if there is a path between any two of its vertices.



connected



not connected

Subgraph: A subgraph H is a graph where $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$

→ If H is a subgraph of G , we write $H \subseteq G$

→ If $H \subseteq G$, then G is said to be a super graph of H .

Isomorphic graphs:

Two graphs G and G' are isomorphic if there is a function $f: V(G) \rightarrow V(G')$ from the vertices of G to the vertices of G' such that

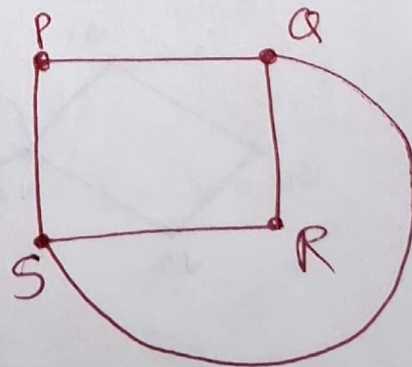
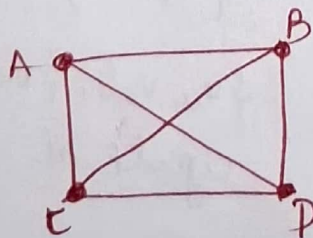
(i) f is one-one

(ii) f is onto

(iii) For each pair of vertices u and v of G

$$\{u, v\} \in E(G) \iff \{f(u), f(v)\} \in E(G')$$

EX: 1. Consider



Consider the following one-to-one correspondence between the vertices of these two graphs.

$$A \longleftrightarrow P, \quad B \longleftrightarrow Q, \quad C \longleftrightarrow R, \quad D \longleftrightarrow S$$

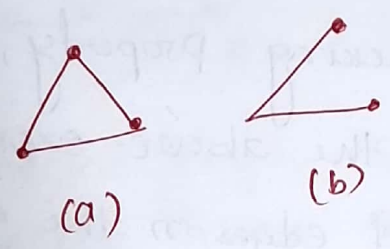
Under this correspondence, the edges in the two graphs correspond with each other as indicated below.

$$\{A, B\} \longleftrightarrow \{P, Q\}, \quad \{A, C\} \longleftrightarrow \{P, R\}, \quad \{A, D\} \longleftrightarrow \{P, S\}$$

$$\{B, C\} \longleftrightarrow \{Q, R\}, \quad \{B, D\} \longleftrightarrow \{Q, S\}, \quad \{C, D\} \longleftrightarrow \{R, S\}$$

we check the above-indicated one-to-one correspondence between the vertices/edges of the two graphs preserves the adjacency of the vertices. The existence of this correspondence prove that two graphs are isomorphic.

EX:



We observe that these two graphs have the same number of vertices but different number of edges. Therefore, although there can exist one-to-one correspondence between the vertices, there cannot be a correspondence between the edges. The two graphs are therefore not isomorphic.

Th ①: The sum of the degrees of all the vertices in a graph is an even number, and this number is equal to twice the no. of edges in the graph.

(or)

For a graph $\sum_{v \in V} \deg(v) = 2|E|$

(or)

In every graph, the no. of vertices of odd degree is even. Euler

Consider a graph with n vertices. Suppose k of these vertices are of odd degree so that remaining $n-k$ vertices are of even degree.

Denote the vertices with odd degree by $v_1, v_2, v_3, \dots, v_k$ and the vertices with even degree $v_{k+1}, v_{k+2}, \dots, v_n$.

Then the sum of the degree of the vertices is

$$\sum_{i=1}^n \deg(v_i) = \sum_{i=1}^k \deg(v_i) + \sum_{i=k+1}^n \deg(v_i)$$

According to hand-shaking property, the sum on the left hand side of the above expression is equal to twice the no. of edges in the graph.

Further the second sum in the right hand side is also even. Therefore the first sum in the right hand side must also be even.

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_k) = \text{even}$$

(or)

In every graph, the no. of vertices of odd degree is even.

Euler

Consider

Consider a graph with n vertices. Suppose k of these vertices are of odd degree so that remaining $n-k$ vertices are of even degree.

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According to hand-shaking property, the sum on the left hand side of the above expression is equal to twice the no. of edges in the graph.

Further the second sum in the right hand side is also even. Therefore the first sum in the right hand side must also be even.

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_k) = \text{even.}$$

$$\sum_{v \in V} \deg(v) = 2|E|$$

Euler circuit and Euler trails.

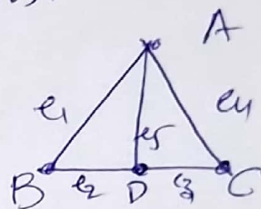
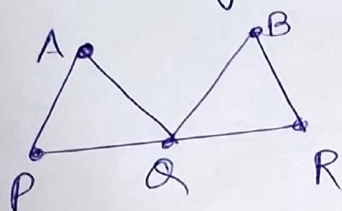
Consider a connected graph G . If there is a circuit in G that contains all edges of G , then that circuit is called an Euler circuit in G .

Note : If there is a trail in G that contains all the edges of G , then that trail is called Euler trail.

Euler graph:

A connected graph that contains Euler circuit is called an Euler graph (Eulerian graph).

ex:



Th: A connected graph G has an Euler circuit iff all vertices of G are of even degree.

First suppose that G has an Euler circuit.

While tracing this circuit we observe that every time the circuit meets a vertex v it goes through two edges incident on v . This is true for all vertices that belong to the circuit. Since the circuit contains all edges, it meets all the vertices at least once. Therefore every vertex is a multiple of two.

Conversely, suppose that all the vertices of G are of even degree.

Prim's Algorithm

①

Input : A connected weighted graph G with n vertices.

Output : A minimal spanning tree T .

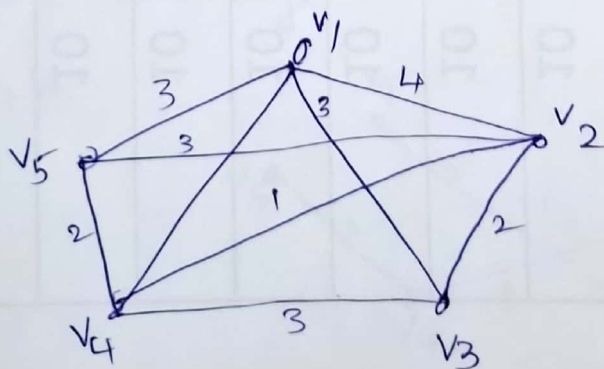
Step 1 : Select an arbitrary vertex v_1 and edge e_1 with minimum weight incident with vertex v_1 .
This forms minimal spanning tree.

Step 2 : If edges $e_1, e_2, \dots, e_i$ have been chosen involving end points $v_1, v_2, \dots, v_{i+1}$

choose an edge $e_{i+1} = v_j v_k$ with $v_j \in T$ and $v_k \notin T$ such that e_{i+1} has smallest weight among the edges of G with precisely one end in $\{v_1, v_2, \dots, v_{i+1}\}$.

Step 3 : Stop after $n-1$ edges have been chosen.

Ex : Find the minimal spanning tree of the weighted graph using prim's algorithm

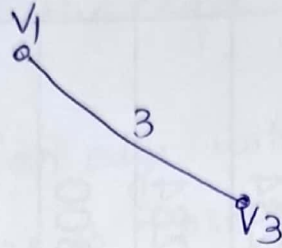


sol → According to Step 1, choose vertex v_1 .

→ Now edge with smallest weight incident on v_1

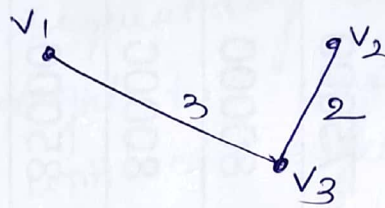
is $v_1 v_3$ or $v_1 v_5$ or $v_1 v_4$

choose $e = v_1 v_3$



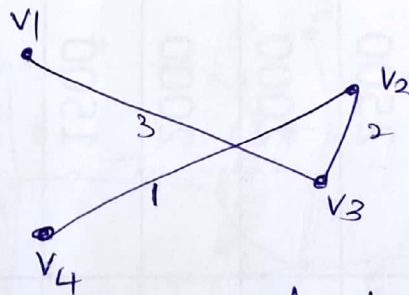
→ choose the edge with smallest weight incident on v_3

$e = v_3 v_2$



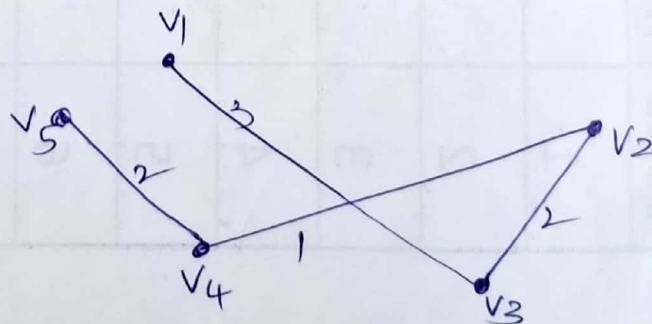
→ choose the edge with smallest weight incident on v_2

$e = v_2 v_4$



→ choose the edge with smallest weight incident on v_4

$e = v_4 v_5$



Requires
MST.

Kruskals Algorithm

Input: A connected weighted graph G with n vertices.

Output: A minimal spanning tree for G .

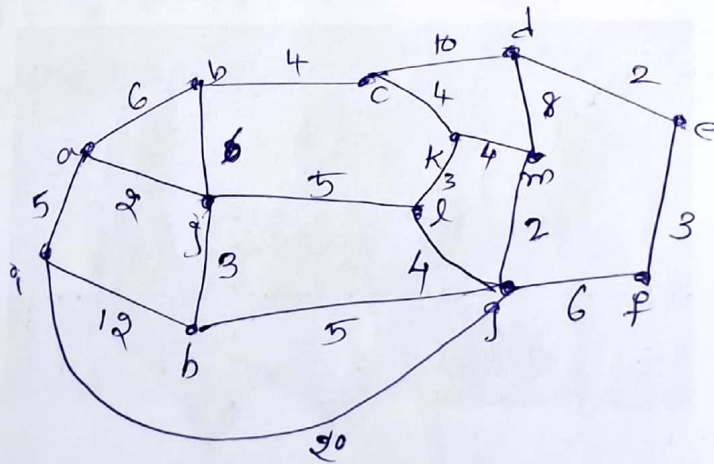
Step 1 → Arrange the edges in order of increasing weights.

Step 2 → select an edge with minimum weight
(If there is more than one edge with minimum weight arbitrarily choose one of these edges.
This is first edge of T

Step 3 → proceed sequentially adding each edge of minimum weight to T which does not result in a cycle until $n-1$ edges are included.

Step 4 → Exit with minimal spanning tree T .

EX:



- Kruskal's algorithm initially picks an edge of lowest weight.
- There are several edges of weight 2, any of them may be selected.
- Suppose we begin with edge $e_1 = aj$
- There still remain edges of weight 2, so choose any other say $e_2 = gm$.
- Among those edges which do not form a cycle with e_1 and e_2 , choose one of the least weight say $e_3 = de$
- Continuing, we may have edges $aj, gm, de, mf, fe, lk, hj, km, ck, bc$.
- Any other edges selected will form a cycle.
- Also $|E| = 12 = |V| - 1 = 13 - 1 = 12$.
- The MST weight is 39

Eulerian graph: An Euler path in a multigraph is a path that includes each edge of the multigraph exactly once and includes or intersects each vertex of the multigraph at least once.

→ A multigraph is said to be transversable if it has an Euler path.

Euler circuit: An Euler circuit is an Euler path whose end points are identical.

That is if an Euler path is a sequence of edges $e_1, e_2, \dots, e_k$ corresponding to the sequence of pairs of vertices $(v_1, v_2), (v_2, v_3), \dots, (v_{k-1}, v_k)$ then e_i 's are all distinct and $v_1 = v_k$.

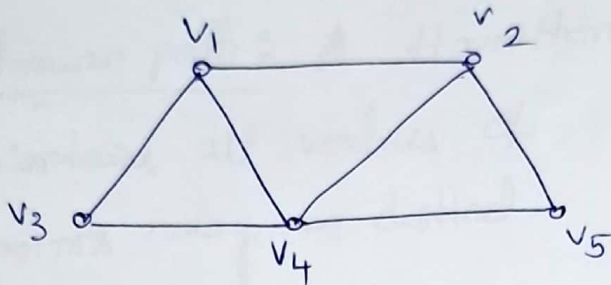
Eulerian multigraph: A multigraph is said to be an Eulerian multigraph if it has an Euler circuit.

Note: 1) A connected graph is semi-Eulerian if and only if it has exactly two vertices of odd degree.

2) A non-directed multigraph has an Eulerian path if and only if it is connected and has 0 or exactly two vertices of odd degree.

3) A nondirected multigraph has an Euler circuit if and only if it is connected and all of its vertices are of even degree. (2)

Ex :



$$\begin{aligned} d(v_1) &= d(v_2) = 3 \\ d(v_3) &= d(v_5) = 2 \\ d(v_4) &= 4 \end{aligned}$$

Clearly given graph has two vertices of odd degree v_1 and v_2 and the remaining vertices are of even degree. Also G is connected.

$v_1 - v_3 - v_4 - v_5 - v_2 - v_4 - v_1 - v_2$ is Euler path.

→ Euler circuit is not there in the graph ; since all the vertices of the graph are not of even degree.

∴ The given graph has Euler path only.

Hamiltonian Graphs :

Hamiltonian cycle : A cycle in a graph G that contains each vertex in G exactly once, except for the starting and ending vertex that appears twice is known as Hamiltonian cycle.

Hamiltonian graph:

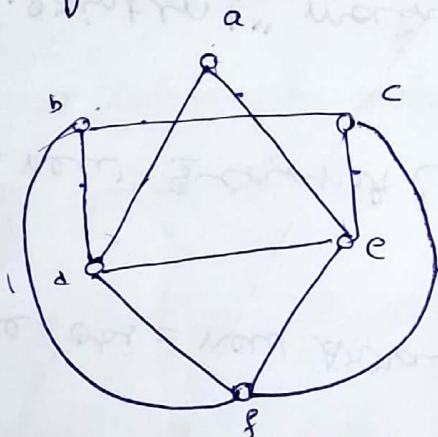
(3)

A Hamiltonian graph is a graph containing a Hamiltonian cycle.

Hamiltonian path: A Hamiltonian path is a simple path that contains all vertices of G but where the end points may be distinct.

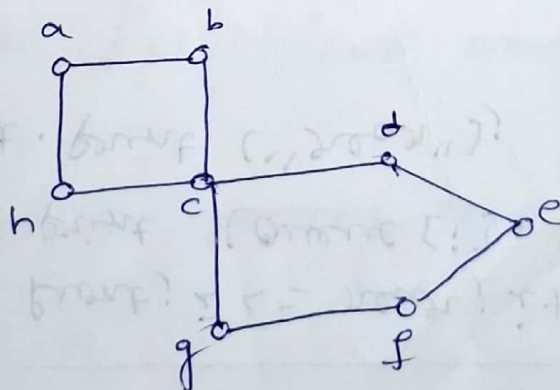
Note:- 1) The complete graph K_n ($n \geq 3$) is Hamiltonian
2) The complete bipartite graph $K_{m,n}$ is Hamiltonian if and only if $m=n$ and $n > 1$.

Ex 1.



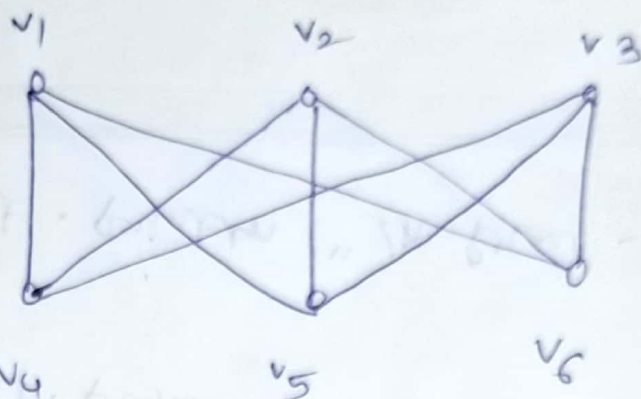
A graph with an EC and HC

Ex 2.



A graph with an EC but no HC

EX: 3



A graph with HC but no EC

Note : A non directed multigraph has an Euler circuit if and only if it is connected and all of its vertices are of even degree.

Spanning tree : A spanning tree of a connected graph G is a subgraph which is a tree and which includes every vertex of G .

Minimal spanning tree : A minimal spanning tree of a connected weighted graph is a spanning tree of least weight that is a spanning tree for which the sum of the weights of all its edges is least among all the spanning trees.

→ Note
 ① The concept of spanning tree exists only for a connected graph G

② If G has n vertices, any spanning tree must necessarily contain $n-1$ edges.